Supplementary material to "Global minimality of controlled-rail brachistochrones in conformal Kerr and Vaidya spacetimes": alternative proofs and computer-assisted steps

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S1. SCOPE

This supplement contains material that the main text uses or announces but does not print in full. References of the form "main text" are to the numbering of the paper; the sections follow its order.

Sections S2–S5 give an alternative proof of the cut-point rule for conformal Schwarzschild (Theorem 2 of the main text) on narrower ranges of q, by convex wavefronts on Hadamard slices in static conformal frames. The convex-front argument is not needed for the main-text cut-point results, whose proofs use the comparison of Lemma V.2 and the count of Proposition III.11. Where it applies, this proof also excludes meetings of distinct null rays in the universal cover of the plane at every swept angle. It works directly on a time interval that need not contain the whole future; finite-window minimality is also established in the main text by Corollary V.7.

The remaining sections contain the computer-assisted steps and the constants of the examples: section S6 the constants of the sharpness example below $q = 3/2$; section S7 the curvature certificates on the equator of Kerr; section S8 the certified critical separations and the Maxwell example of the proper-time branch; section S9 the examples with a time-dependent factor; section S10 the proofs with constants in the example on the equator with spin; and section S11 the constants of the Vaidya example. In each case the main text states the result and the structure of its proof. Scripts, logs and certificate files are in the accompanying repository.

S2. STATIC CONFORMAL FRAMES

In the plane the auxiliary spacetime is three-dimensional. Its Weyl tensor vanishes and the screen of a null geodesic is one-dimensional, so the in-plane Jacobi equation is

$$y'' + \mathrm{Ric}(k,k)\, y = 0 \tag{1}$$

in an affine parameter. Null conjugate points are conformally invariant, and we use conformal frames in which the slices carry a *static* metric. For a constant $b \in [3/2, \infty]$ let

$$N_b = \Big(1 - \frac{f}{b}\Big)^{-1} \ (N_\infty = 1), \qquad \bar{g}_b = \frac{N_b}{N}\, g_{\mathrm{opt}} = -N_b V^2\, \mathrm{d}\eta^2 + N_b\, h_0. \tag{2}$$

The frame $\bar{g}_b$ does not freeze A: it is another conformal representative of the same null cone, with slice metric $h_b = N_b h_0$ independent of η.

Lemma S2.1 (no time derivatives)**.** *For a metric $-\tilde{V}(\eta,x)^2\mathrm{d}\eta^2 + \tilde{h}(x)$ with $\tilde{h}$ independent of η, and $k = (1, \tilde{V}e)$ with $|e|_{\tilde{h}} = 1$,*

$$\mathrm{Ric}(k,k) = \tilde{V}^2 K_{\tilde{h}} + \tilde{V}\, \mathrm{Hess}_{\tilde{h}} \tilde{V}(e_\perp, e_\perp), \tag{3}$$

with $e_\perp$ the $\tilde{h}$-unit vector orthogonal to e. In particular $\mathrm{Ric}_{\bar{g}_b}(k,k)$ depends on η only through the value $q(\eta)$, not through A' or A''.

Proof. For such a metric $R_{00} = \tilde{V}\Delta_{\tilde{h}}\tilde{V}$, $R_{0i} = 0$ and $R_{ij} = R_{\tilde{h},ij} - \tilde{V}^{-1}\mathrm{Hess}_{\tilde{h}}\tilde{V}_{ij}$: the time derivatives of $\tilde{V}$ cancel. In two spatial dimensions $R_{\tilde{h}} = K_{\tilde{h}}\tilde{h}$, and contracting with k gives (3). For $\bar{g}_b$, $\tilde{V}^2 = N_b(1 - f/q)$ and $\tilde{h} = N_b h_0$. □

The normalization $k^\eta = 1$ is not affine; for an affine tangent the contraction is multiplied by $(\mathrm{d}\eta/\mathrm{d}\lambda)^2 > 0$, which does not change its sign. Writing $e = \cos\alpha\, e_r + \sin\alpha\, e_\varphi$ in an h_0-orthonormal frame, $\mathrm{Ric}_{\bar{g}_b}(k,k) = C_r\cos^2\alpha + C_\varphi\sin^2\alpha$.

Proposition S2.2 (sign certificates)**.** *Write $R = r/M = 2+x$, $x > 0$. Then $\mathrm{Ric}_{\bar{g}_b}(k,k) < 0$ for every null direction and every $r > 2M$ in each of the following cases.*

(1) $b = \infty$ and $q \geq 2$. Here $C_r = -M\{Mq + (2q-1)(r-2M)\}/(qr^4)$ and $C_\varphi = -MP/\{qr^5[2M + (q-1)r]\}$, where, with $q = 2 + d$, $d \geq 0$,

$$M^{-3}P = (2d^2+2d)x^3 + (9d^2+24d+12)x^2 + (12d^2+48d+48)x + 4d^2 + 16d + 16.$$

(2) $b = 2$ and $q \geq 5/3$. Here $C_r = -P_r/[M^2qR^5(R+2)^2]$, $C_\varphi = -P_\varphi/[M^2qR^5(R+2)((q-1)R+2)]$ with

$$\begin{aligned} P_r &= x^4 + (12q-4)x^3 + (72q-36)x^2 + (112q-32)x + 32q, \\ P_\varphi &= (3q-5)(q-1)x^4 + (20q^2-37q+14)x^3 \\ &\quad + (48q^2-42q)x^2 + 48q^2x + 16q^2. \end{aligned}$$

(3) $b = 5/3$ and $11/7 \leq q \leq 2$. Here $C_r = -Q_r/[4M^2qR^5(R+3)^2]$, $C_\varphi = -Q_\varphi/[2M^2qR^5(R+3)((q-1)R+2)]$ with

$$\begin{aligned} Q_r &= 4(2-q)x^4 + (37q-3)x^3 + (390q-210)x^2 + (700q-200)x + 200q, \\ Q_\varphi &= (7q-11)(q-1)x^4 + (47q^2-87q+34)x^3 \\ &\quad + (114q^2-102q)x^2 + 116q^2x + 40q^2. \end{aligned}$$

(4) q constant, $b = q \geq 3/2$ (ultrastatic frame). Here $\mathrm{Ric}(k,k) = K_t$, the optical Gauss curvature of Paper II, and with $q = 3/2 + d$ the numerator of $-K_t$ is

$$2d^2x^3 + 9d^2x^2 + 12d^2x + 4d^2 + dx^3 + 16dx^2 + 34dx + 12d + \tfrac{23}{4}x^2 + 24x + 9.$$

In each case all coefficients are non-negative and the constant term is positive on the stated range of q (for $20q^2 - 37q + 14$ and $47q^2 - 87q + 34$, check the lower end of the range and monotonicity).

Proof. Direct computation from (3), verified symbolically by the scripts `supplement_*.wls` of the accompanying repository (the four cases and the frames; the file `RESULTS_MAP.md` there gives the script of each step). □

Remark S2.3 (sharpness of the frames)**.** The certificates are frame-dependent sufficient conditions, not thresholds for minimality. In the frame $b = \infty$, $\lim_{r\to\infty} r^3 C_\varphi = -2M(q-2)/q$, so the sign fails far out for $q < 2$; in the frame $b = 2$ it fails far out for q just below $5/3$, and in the frame $b = 5/3$ for q just below $11/7$. For constant q the threshold $3/2$ of item 4 is exactly that of Paper II. The frame must be fixed along the whole extremal: switching b in time would destroy the static slice metric that the next two lemmas use.

Lemma S2.4 (Hadamard slices)**.** *For $b \in [3/2, \infty]$ the universal cover of $(\{r > 2M\} \times S^1, h_b)$, with the angle unwrapped, is a Hadamard surface: complete, simply connected, with $K_{h_b} < 0$.*

Proof. K_{h_b} is the optical curvature K_t at constant $q = b$, negative by Proposition S2.2(4); for $b = \infty$, $K_{h_0} = M(3M-2r)/r^4 < 0$. The radial integrand $\sqrt{N_b}/f$ behaves as $2M/(r-2M)$ at the inner end and tends to a positive constant at infinity, so radial distances diverge at both ends; and $N_b \geq 1$ with $r^2/f - 27M^2 = (r-3M)^2(r+6M)/(r-2M) \geq 0$ bounds the angular radius below by $3\sqrt{3}M$. A curve of finite length therefore escapes neither radially nor in the unwrapped angle. □

S3. POSITIVE EXPANSION, CONVEX FRONTS, NO CROSSING

Fix a frame $\bar{g}_b$ in which Proposition S2.2 applies along the time interval considered. Let $F_t : S^1 \to \tilde{S}_t$ be the *front*: $\alpha \mapsto$ the position at $\eta = t$ of the null geodesic from the (lifted) launch event $\tilde{o}$ with initial direction α, in the universal cover $\tilde{S}$ of the slice.

Lemma S3.1 (expansion)**.** *Along every null geodesic of the cone of $\tilde{o}$, the in-plane Jacobi field y with $y(0) = 0$, $y'(0) > 0$ satisfies $y \geq y'(0)\lambda$ and $\theta = y'/y \geq 1/\lambda > 0$.*

Proof. While $y > 0$, $y'' = -\mathrm{Ric}(k,k)y \geq 0$, so $y' \geq y'(0)$ and $y \geq y'(0)\lambda$: there is no first zero. Moreover $(\lambda y' - y)' = \lambda y'' \geq 0$, so $\lambda y' \geq y$. □

Lemma S3.2 (convex fronts)**.** *Each F_t is an immersion, it moves along its h_b-normal with speed $\tilde{V} > 0$, and its geodesic curvature κ with respect to the direction of motion is positive (with this convention a circle has $\kappa > 0$ with respect to its outward normal).*

Proof. The variation at fixed η is $Y + c_1 k$, with Y the Jacobi field of the cone; since $\bar{g}_b(Y,k) = 0$, $|\partial_\alpha F_t|^2_{h_b} = \bar{g}_b(Y,Y) = y^2 > 0$. The spatial velocity v of a ray satisfies $h_b(v, \partial_\alpha F_t) = \bar{g}_b((0, \partial_\alpha F_t), k) = 0$ and $|v|_{h_b} = \tilde{V}$. For a normal flow in a static metric, $\partial_t \log |\partial_\alpha F_t|_{h_b} = \tilde{V}\kappa$, and the left side is $\theta/(\mathrm{d}\eta/\mathrm{d}\lambda) > 0$ by Lemma S3.1. □

Lemma S3.3 (no crossing in the universal cover)**.** *Two distinct null geodesics from $\tilde{o}$ do not meet.*

Proof. On the cover every slice is a Cauchy surface: causal curves have $|\mathrm{d}x/\mathrm{d}\eta|_{h_b} \leq \tilde{V}$, bounded on compact time intervals, and h_b is complete (Lemma S2.4). Let $R_t = J^+(\tilde{o}) \cap \tilde{S}_t$: compact, with interior containing the launch point, non-decreasing in t (the lines L_x are timelike), and $\partial R_t \subset F_t(S^1)$ because $\partial J^+(\tilde{o})$ is generated by null geodesics from $\tilde{o}$.

Suppose a crossing exists and let t_* be the first time at which F_t fails to be injective; for t larger than the clock of $\tilde{o}$ and sufficiently close to it, the front is the section at clock t of the null cone of $\tilde{o}$ in a normal neighborhood, an embedded circle, so t_* is larger than that clock; it is attained with two distinct directions, because the immersion is uniform on compact sets. For $t < t_*$, F_t is a Jordan curve. Since $\partial R_t \subset F_t(S^1)$, the compact set R_t contains no point of the unbounded component of the complement, and it contains the whole bounded component, which is connected and meets the interior of R_t at the launch point; so R_t is the closed disk bounded by F_t, with outward direction of motion. By Lemma S3.2 its boundary has positive curvature, so R_t is locally geodesically convex, hence convex on the Hadamard surface: a shortest curve inside the disk between two of its points is an ambient geodesic, hence the unique segment.

The limit body $R_{t_*} = \overline{\bigcup_{t<t_*} R_t}$ is compact and convex with non-empty interior, and $F_{t_*}(S^1) = \partial R_{t_*}$: a limit of earlier boundary points that were interior to the limit body would lie inside a small geodesic triangle with vertices in the union, hence inside some R_s, $s < t_*$, by convexity. The boundary of a compact convex body with interior in a Hadamard surface is a topological circle; F_{t_*} is an immersion onto it, hence a covering, and its degree about the launch point is one, as for the earlier simple fronts. A covering of degree one is injective, contradicting the choice of t_*. □

S4. THE THEOREM IN STATIC FRAMES

Proposition S4.1 (convex fronts)**.** *Assume (H1) and one of the following on the whole of I:*

$$\text{(a) } q(\eta) \geq 2; \qquad \text{(b) } q(\eta) \geq \tfrac{5}{3}; \qquad \text{(c) } \tfrac{11}{7} \leq q(\eta) \leq 2; \qquad \text{(d) } A \text{ constant and } q \geq \tfrac{3}{2}. \quad (4)$$

Then the fronts of section S3 are convex in a static frame, two distinct null geodesics from $\tilde{o}$ never meet in the universal cover (Lemma S3.3), and the same minimality and cut-point

conclusions as in Theorem 2 hold throughout I, by the argument below; here I need not contain the whole future.

Proof. Choose the frame $b = \infty, 2, 5/3$ or q in cases (a)–(d). By Proposition S2.2 and Lemma S3.1 there is no in-plane conjugate point, and by Lemma V.1 the first out-of-plane one is at $\Delta\varphi = \pi$. Suppose γ had a cut point before π. It is not conjugate, so by Fact III.4(4) a second maximal null geodesic σ from o reaches the same event. Being maximal, σ has swept angle at most π. For $0 < \Delta\varphi < \pi$ the center, the launch point and the end point fix a unique plane; the opposite sense would need $2\pi - \Delta\varphi > \pi$, and further windings more. So σ lies in the plane of γ, with the same sense and the same unwrapped angle, and γ, σ lift to null geodesics from $\tilde{o}$ meeting at the same point of the cover, which Lemma S3.3 excludes. Hence every event before π is on $\partial J^+(o)$, reached by a unique null segment, and Theorem 1 gives items 1–2. At $\Delta\varphi = \pi$ the event stays on the causal boundary by continuity ($I^+(o)$ is open), and the rotations of Lemma V.1 give a family of distinct minimizers: it is the first cut point. Beyond it the conjugate point is interior, the segment is no longer achronal, and an earlier arrival exists.

For a radial extremal the rotations fix it and the argument above does not apply. Every causal competitor obeys $|\mathrm{d}r/\mathrm{d}\eta| \le fV$, while the radial extremals solve $r' = \pm fV$. For the outgoing branch $w = R - r$ satisfies $w' = a(\eta)w + d(\eta)$ with a bounded on the compact set traversed and $d = fV - r' \ge 0$; an integrating factor gives $w \ge 0$, with $w > 0$ at arrival unless $d = 0$ almost everywhere, which forces saturation of the radial speed and zero angular velocity. The ingoing branch is analogous, and radial completeness prevents reaching $r = 2M$ or infinity in finite time. □

Lemma S3.3 excludes meetings of distinct rays in the cover at every swept angle, not only below π; the uniqueness proved in section V C is the weaker statement needed for Theorem 2.

S5. THE WHOLE FAMILY OF STATIC FRAMES

Cases (a)–(c) of Proposition S4.1 are three frames of a one-parameter family. For a constant $b > 3/2$, write the negatives of the numerators of C_r and C_φ in the frame $\bar{g}_b$ (with $M = 1$, $r = 2 + x$; the denominators $qr^5(2 + (b-1)r)^2$ and $qr^5(2 + (b-1)r)(2 + (q-1)r)$ are positive):

$$\begin{aligned} P_r &= (b-1)(2bq - b - 4q + 3)x^4 + (13b^2q - 6b^2 - 24bq + 11b + 8q - 2)x^3 \\ &\quad + 6b(5bq - 2b - 4q + 1)x^2 + 4b^2(7q - 2)x + 8b^2q, \\ P_\varphi &= (q-1)(2bq - 4b - q + 3)x^4 + (13bq^2 - 24bq + 8b - 6q^2 + 11q - 2)x^3 \\ &\quad + 6q(5bq - 4b - 2q + 1)x^2 + 4q^2(7b - 2)x + 8bq^2. \end{aligned}$$

Proposition S5.1 (the band of a frame)**.** *For $b, q \ge 3/2$ the coefficients of $x^0, \dots, x^3$ in P_r and P_φ are positive. The coefficients of x^4 are non-negative exactly when*

$$q_-(b) := \frac{4b-3}{2b-1} \;\le\; q \;\le\; q_+(b) := \begin{cases} \dfrac{3-b}{4-2b}, & b < 2, \\ +\infty, & b \ge 2. \end{cases} \tag{5}$$

Hence $\mathrm{Ric}_{\bar{g}_b}(k,k) < 0$ for every $r > 2M$ and every null direction if and only if q lies in (5)*; outside it the sign fails at large r.*

Proof. With $b = 3/2 + u$, $q = 3/2 + v$ each coefficient of degree 0–3 is a polynomial in u, v with positive coefficients; for instance the radial x^3 coefficient is $13u^2v + \frac{27}{2}u^2 + 15uv + \frac{31}{2}u + \frac{5}{4}v + \frac{23}{8}$, and the tangential one follows by exchanging u and v. The x^4 coefficients are non-negative iff $(2b-4)q + 3 - b \ge 0$ and $(2b-1)q + 3 - 4b \ge 0$; for $b \ge 2$ the first is automatic, since at $q = q_-(b)$ it equals $3(b-1)(2b-3)/(2b-1) > 0$. When an x^4 coefficient vanishes the

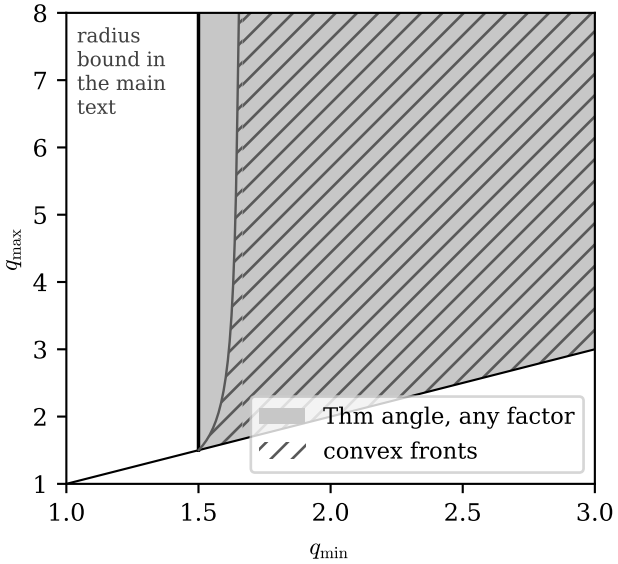


FIG. 1. Arrival branch, $a = 0$, in the $(q_{\min}, q_{\max})$ plane: Theorem 2 of the main text covers $q_{\min} \geq 3/2$ for an arbitrary smooth factor (grey); hatched, the narrower ranges where the convex-front argument of Propositions S4.1 and S5.2 also applies.

constant term is still positive. The asymptotics $\lim_{r\to\infty} r^3 C_r = -(3-b-4q+2bq)/((b-1)q)$ and $\lim_{r\to\infty} r^3 C_\varphi = -(3-4b-q+2bq)/((b-1)q)$ give the converse. □

The bands are exact for this criterion and this family of frames, not thresholds for minimality. Since $q'_\pm(b) = 2/(2b-1)^2,\ 2/(4-2b)^2 > 0$, the best frame for a range $[q_{\min}, q_{\max}]$ is the largest b with $q_-(b) \leq q_{\min}$, namely $b_* = (3-q_{\min})/(4-2q_{\min})$ for $q_{\min} < 5/3$, and $q_+(b_*) = (9-5q_{\min})/(10-6q_{\min})$.

Proposition S5.2 (continuous family of frames)**.** *Assume (H1), and let* $q_{\min} = \inf_I q$, $q_{\max} = \sup_I q$. *If*

$$q_{\min} \geq \tfrac{5}{3}, \qquad \text{or} \qquad \tfrac{3}{2} < q_{\min} < \tfrac{5}{3} \ \text{ and } \ q_{\max} \leq \frac{9-5q_{\min}}{10-6q_{\min}},$$

then $\mathrm{Ric}(k,k) < 0$ *in a static frame, the fronts are convex, and two distinct null geodesics from* $\tilde{o}$ *never meet in the universal cover, for an arbitrary* C^2 *factor.*

Proof. Use the frame $\bar{g}_{b_*}$ ($b = 2$ if $q_{\min} \geq 5/3$): Proposition S5.1 gives $\mathrm{Ric}(k,k) < 0$, Lemma S2.4 gives Hadamard slices, and Lemmas S3.1–S3.3 apply verbatim. □

Proposition S5.2 contains cases (a)–(c), and every $q_0 > 3/2$ has a non-constant band around it: $q_{\min} = 1.55$ allows $q_{\max} \leq 1.786$, $q_{\min} = 1.6$ allows $q_{\max} \leq 2.5$, $q_{\min} = 1.65$ allows $q_{\max} \leq 7.5$. A single frame must work on the whole time interval.

S6. CONSTANTS IN THE SHARPNESS EXAMPLE

This section verifies the constants stated for $q = 6/5$ in Proposition V.8. Here $A_1 = -3\sqrt{6}$ and, with $x = M/r$, $F = \sqrt{1-2x}\,(1+10x)/(1+12x-40x^2)$ in (19). The substitution $\sqrt{1-2x} = (p^2-12p+30)/(p^2-10p+30)$, $\sqrt{1+10x} = (30-p^2)/(p^2-10p+30)$, $0 \leq p \leq 11/10$, covers $r \geq 10M$ monotonically and makes $\rho^2 G/M^2$ and $-\rho^3 G'/M^2$ rational in p. The non-negativity of $\rho^2 G$, $465M^2 - \rho^2 G$, $-\rho^3 G'$ and $930M^2 + \rho^3 G'$ on that interval follows from the signs of the Bernstein coefficients of their numerators and denominators,

computed in exact rational arithmetic. So (20) holds with $K = 465$ and $\rho_* = \rho(10M)$. With $D \geq D_0 = 2 \times 10^4 M$ and $b_\pm = \beta_\pm \sqrt{D/D_0}$, $\beta_- = 300M$, $\beta_+ = 2000M$, put $t = D/D_0 \geq 1$, so that $b_\pm^2/D^2 = \beta_\pm^2/(D_0^2 t) \leq \epsilon_\pm$ with $\epsilon_- = 9/40000$ and $\epsilon_+ = 1/100$. Since $D_0 > KM/c$, $0 < F_{\min} := 1 - cM/D_0 - KM^2/D_0^2 \leq F(D) \leq 1$. Moreover $\rho_* = 5\sqrt{15}\, M < 20M$ and $b_c \leq \rho(3M) = \sqrt{486/13}\, M < 7M$, so $b_\pm \geq 300M > \max(\rho_*, b_c)$ and the bounds of the proof of Proposition V.8 apply. Write $s_\pm(t) = \sqrt{1 - \epsilon_\pm/t}$. For the lower bound, $F(D) \leq 1$, $\sqrt{1-\epsilon_-} \leq s_-(t) \leq 1$ and $D|R_1| \leq \pi K M^2 D_0/(\beta_-^3 \sqrt{t})$ are used; for the upper bound, $F(D)/s_+(t) \geq F_{\min}$, $s_+(t) \leq 1$ and $t^{-1/2} \leq 1$. With $c = 3\sqrt{6}$ this gives

$$D\Phi_b(b_-) \geq -\frac{2}{\sqrt{1-\epsilon_-}} + \frac{2cMD_0\sqrt{1-\epsilon_-}}{\beta_-^2} - \frac{\pi K M^2 D_0}{\beta_-^3} > 0.1832,$$

$$D\Phi_b(b_+) \leq -2\Big(1 - \frac{cM}{D_0} - \frac{KM^2}{D_0^2}\Big) + \frac{2cMD_0}{\beta_+^2} + \frac{\pi K M^2 D_0}{\beta_+^3} < -1.922.$$

Moreover $b_+ \leq D/2$ for $t \geq 1$ and $b_- \geq 300M > \pi KM/(2\sqrt{3}\, c)$, so $\Phi < \pi$ on $[b_-, b_+]$; and $\rho(8200M)^2 = 6 \cdot 8200^4 M^2/(8198 \cdot 8210) > D_0^2$.

S7. CURVATURE CERTIFICATES ON THE EQUATOR

This section describes the certificates behind Propositions IX.6 and IX.7 and the radial confinement of the rays in case (2) of the latter. Each certificate is a closed algebraic lemma: a sign condition for a polynomial with rational coefficients on a box $[0,1]^d$, together with its proof. The polynomials are, up to an explicit positive factor, the geometric inequalities of the main text written in the box variables; this identification is a separate symbolic computation (see the last paragraph).

a. Positivity on a cell. On a cell of a binary subdivision of $[0,1]^d$, rescaled to the unit box, a polynomial is written in the tensor Bernstein basis of its multidegree. If all Bernstein coefficients are positive (non-negative), the polynomial is positive (non-negative) on the cell. A certificate of this kind is a subdivision tree, read in preorder, whose nodes are bisections along a named axis and whose leaves are cells where the criterion holds. The leaves cover the box by construction.

b. Inequalities with one square root. Several inequalities have the form $P_0 + P_1\sqrt{g} \geq 0$ with $g \geq 0$. On a leaf, let g_0 and g_1 be the smallest and the largest Bernstein coefficient of g (with g_0 replaced by 0 if negative), so that $g_0 \leq g \leq g_1$ there. Two affine bounds for $v = \sqrt{g}$ are used: the chord through (g_0, s_0) and (g_1, s_1), with rationals $s_0^2 \leq g_0$, $s_1^2 \leq g_1$, which lies below $\sqrt{g}$ by concavity, and the tangent bound $\sqrt{g} \leq \sigma/2 + g/(2\sigma)$ for a rational $\sigma > 0$. Since $P_0 + P_1 v$ is affine in v, it suffices that $P_0 + AP_1 + BP_1 g$ has non-negative Bernstein coefficients for the two pairs (A, B) given by these lines. A second form avoids the envelopes: with $Q = P_1^2 g - P_0^2$, on each leaf one of the sets $\{P_0 \geq 0, P_1 \geq 0\}$, $\{P_1 \geq 0, Q \geq 0\}$, $\{P_0 \geq 0, -Q \geq 0\}$ is certified, and each implies the inequality.

c. The certificates. Table I lists them, 33 in all. The box variables are affine images of the physical ones, except where stated: $u = 1/q$, $t = \tan(\alpha/2)$ for the direction of the null vector, and $R = r_+/M$ with $a^2 = R(2-R)$. The range $t \in [-1, 1]$ covers $\cos\alpha \geq 0$; the other sign is covered because the remaining terms are even in $\cos\alpha$, while the term linear in it is bounded with both signs (I1$_\pm$) in group (3) and enters through its square in group (4).

(1) *Static part, constant factor* (used in Proposition IX.7(1) for $R_{\text{stat}} < 0$): box (X, U, Z) with $x = 2M/r = X$, $u = U/2$, $a^2/M^2 = \frac{1089}{2500} Z$, that is $q \geq 2$ and $|a| \leq 33M/50$. Four non-negativity lemmas, one cell each; together they give $U \leq 0$, $V \leq 0$, $V^2 \geq J^2$ and $J^2 \leq 4U^2$ for $\text{Ric} = U(1-s^2) + Vs^2 + Js$. The inequalities are strict for $0 < X < 1$: the Bernstein basis functions in X are then positive, and in the certificate polynomials for $-U$, $-V$, $V^2 - J^2$ and $4U^2 - J^2$ (up to their positive factors; files

`group1_minus_nU`, `group1_minus_nV`, `group1_P3`, `group1_P6`) the slices of X-index 1, 1, 2 and 1 have positive coefficients (smallest 4, 2, 56733/27500 and 96); in particular $U < 0$ and $V < -|J|$. With $t = |s|$, $|J| \le -2U$ gives $Vt^2 + |J|t \le -2Ut(1-t)$, hence $\mathrm{Ric} \le U(1-t)^2 < 0$ for $t < 1$, and $\mathrm{Ric} \le V + |J| < 0$ for $t = 1$.

(2) *Proposition IX.6*: box (X, R_v, U) with $x = r_+/r = X$, $R = 1 + R_v$, $u = \frac{3}{5}U$. One certificate with three sufficient sets on its 19 leaves: $\{V \le 0,\ V^2 \ge J^2\}$ together with one of $J^2 \le -4U(U-V)$ (the value at the vertex is non-positive), $V \ge U$ (the quadratic in s is convex), or $\{U \le 0,\ J^2 \le U^2\}$. Each set implies $\max_s \mathrm{Ric} \le 0$; this implication is part of the proof, not of the certificate.

(3) *Proposition IX.7(1), the jets*: box (Y, T, U, A) with $y = \sqrt{1 - 2M/r} = Y$, $t = 2T - 1$, $u = \frac{2}{5}U$, $a/M = \frac{33}{50}A$. Three inequalities, $|D_1| \le \frac{5}{4}(-R_{\mathrm{stat}})$ with either sign of the D_1 term ($\mathrm{I1}_\pm$) and $|D_2| \le \frac{1}{3}(-R_{\mathrm{stat}})$ (I2), each of the form $P_0 + P_1\sqrt{g} \ge 0$. The main certificate of each delegates two corners where both y and either $1 - T$ or a are small: the corner C1 ($y \le 1/8$, $T \ge 15/16$) is the affine image of a separate box; the corner C2 ($y \le 1/8$, $A \le 1/8$) is covered by two sectors, C2A ($a \le 2y$, with $a = 2yA'$) and C2B ($2y \le a$, with $y = aY'/2$), in which the common power of the small parameter is divided out of P_0 and P_1.

(4) *Proposition IX.7(2)*: for each band, box (X, R_v, U, T) with $x = r_+/r = x_b X$, $x_b = 10R_{\mathrm{lo}}/(11R_{\mathrm{lo}} - 1)$, $R = R_{\mathrm{lo}} + (R_{\mathrm{hi}} - R_{\mathrm{lo}})R_v$, $u = \frac{2}{5}U$, $t = 2T - 1$. Three inequalities with one square root: $-R_\sigma \ge 0$ (S), $K_2(-R_\sigma) - Cc^2 \ge 0$ (I2) and $K_1^2 R_\sigma^2 - D_1^2 \ge 0$ (I1).

(5) *Radial confinement in case (2)*: box (Y, R_v, U) with $r = R + y(R-1)/10$, $R = \frac{57}{50} + \frac{31}{50}R_v$, $u = \frac{2}{5}U$. Writing $\partial_r G(X_\pm, X_\pm) = \alpha \pm \beta\sqrt{N\Delta}$, four positivity lemmas give the signs of the two denominators, $-\alpha > 0$ and $\alpha^2 - \beta^2 N\Delta > 0$, hence $\partial_r G(X_\pm, X_\pm) < 0$ on $(r_+, r_+ + (r_+ - 1)/10]$.

TABLE I. The curvature certificates on the equator: number of leaves of the subdivision and largest degree in each box variable. For group (3), the main certificate (with the number of leaves delegated to the corners) and, in brackets, the corner certificates C1, C2A, C2B.

group	certificates	leaves; degrees
(1)	four lemmas	1 each; up to $(12, 6, 2)$
(2)	K_∞	19; $(16, 12, 8)$
(3)	$\mathrm{I1}_+$	2007 (+144 delegated) [1, 4, 4]; $(22, 4, 4, 6)$ in the main box
	$\mathrm{I1}_-$	2024 (+144) [1, 4, 4]
	I2	2229 (+136) [1, 1, 1]
(4)	B1: S, I2, I1	2, 20, 48
	B2	6, 36, 58
	B3	20, 66, 238
	B4	81, 148, 1414; I1 of degree $(21, 17, 8, 8)$
(5)	four lemmas	1 each; up to $(22, 22, 6)$

d. The verifier. The certificates are distributed as portable files, with polynomials given by their monomials and rational coefficients in exact form, and are checked by an independent verifier that imports none of the code that produced them and uses different algorithms: Bernstein coefficients from the explicit formula $b_I = \sum_{J \le I} a_J \prod_k \binom{i_k}{j_k} / \binom{n_k}{j_k}$ instead of matrix products, and halving of a cell by direct binomial sums instead of de Casteljau's scheme. It recomputes every product $P_1 g$, checks $g \ge 0$ at the root, the shape of the tree, the envelope data $s_0^2 \le g_0$, $s_1^2 \le g_1$, $\sigma > 0$ on every leaf and, for the corners

of group (3), the exact identities $P_0(\varphi) = mP_0^{\mathrm{del}}$, $\lambda P_1(\varphi) = mP_1^{\mathrm{del}}$, $g(\varphi) = \lambda^2 g^{\mathrm{del}}$ with positive monomials m, λ, and that delegation is acyclic. All 33 certificates pass in one run of about twenty minutes, most of it on B4, I1. Fifteen deliberate alterations (a perturbed polynomial, a false envelope, a truncated tree, a split replaced by a leaf, a sign change of g, an empty sufficient set, circular delegation and others) are all rejected, each for the expected reason.

e. What the verifier does not check. Four steps are mathematical and are proved separately. (i) The identification of each polynomial with its geometric inequality, up to a positive factor whose sign is checked factor by factor; this is an exact symbolic identity, reconstructed independently from the curvature of h, of $\bar{g}_\infty$ and of the weighted representative. (ii) The implications from the sufficient sets of group (2) to $\max_s \mathrm{Ric} \le 0$, including degenerate cases, and the analogous deduction for group (1). (iii) The covering of C2 by the two sectors: a point with $y \le 1/8$ and $a \le 33/400$ has either $a \le 2y$ or $2y \le a$, and the case $y = a = 0$ follows by continuity. (iv) Continuity at the boundary points where the monomial m vanishes: the composed expression is polynomial in the box variables and in the square root, hence continuous there.

S8. THE CERTIFIED CRITICAL SEPARATIONS AND THE MAXWELL EXAMPLE

This section contains the computer-assisted parts of Propositions X.6 and X.8; the reduction to finitely many sign conditions is in the main text.

A. The certified grid

Computer-assisted part of the proof of Proposition X.6. All enclosures use outward-rounded interval arithmetic, with derivatives in t computed exactly by second-order Taylor jets (no finite differences). Since angles and the number of maxima do not depend on A, we set $M = A = 1$. The parameter q enters as an interval: for $q = 6/5$ one that contains both $6/5$ and its binary64 approximation, so that the statement holds for the rational value. Write Δ and ℓ with the regular kernel: $r = t + (r_0 - t)u^2$, $u \in [0, 1]$, and $\rho^2(r) - \rho^2(t) = (r - t)W/D$ with $W = (q - 1)rt(r + t - 2) + 2(r^2 + rt + t^2 - 2r - 2t)$ and $D = ((q - 1)r + 2)((q - 1)t + 2)$, so that the integrands are smooth on $[0, 1]$. Each integral in u is enclosed cell by cell, either by $|U|\,F(U)$ or by the midpoint rule with Lagrange remainder, $\int_U F \in F(m)|U| + F''(U)|U|^3/24$, where F'' is the second u-derivative enclosed by nested jets.

Ends. On $t \in (2M, t_{\mathrm{lo}}]$ and $[t_{\mathrm{hi}}, r_0)$, with $t_{\mathrm{lo,hi}}$ at relative distance 10^{-4} from the ends, we use the variable ρ and $\Delta'(L) = 2\int_1^{\rho_0/L} w'(Lx)\,\mathrm{d}x/\sqrt{x^2 - 1} - 2w(\rho_0)/\sqrt{\rho_0^2 - L^2}$, $w = \mathrm{d}R/\mathrm{d}\rho$, enclosed in the variable $r = 2 + s^2$ on contiguous cells whose last edge is the enclosure of $s_0 = \sqrt{r_0 - 2}$ itself. With $|w'| \le B$ on $[0, s_0]$,

$$\tfrac{1}{2}\Delta'(L) \le B\,\mathrm{arcosh}(\rho_0/L) - \frac{w(\rho_0)}{\sqrt{\rho_0^2 - L^2}},$$

decreasing in L, so its sign at $L = \rho(t_{\mathrm{hi}})$ gives $\Delta' < 0$ on $[t_{\mathrm{hi}}, r_0)$. Near $2M$ a lower bound by pieces, $m_k \le w'$ on each cell, each piece weighted by the extreme value of $\mathrm{arcosh}(\rho_{k+1}/L) - \mathrm{arcosh}(\rho_k/L)$ over $L \in (0, \rho(t_{\mathrm{lo}})]$, gives $\Delta' > 0$ on $(2M, t_{\mathrm{lo}}]$.

Interior. $[t_{\mathrm{lo}}, t_{\mathrm{hi}}]$ is bisected until each piece carries one of three certified labels: $\Delta' > 0$ (P), $\Delta'' < 0$ (C), $\Delta' < 0$ (N). The pieces tile $[t_{\mathrm{lo}}, t_{\mathrm{hi}}]$, adjacent pieces sharing exact binary end points. Here primes are derivatives in t; since $\mathrm{d}L/\mathrm{d}t > 0$, the signs of Δ', its zeros and their non-degeneracy are the same in L. In every case there is a C piece and no N piece precedes a P piece; a P piece cannot be adjacent to an N piece, since Δ' is continuous. On a run of C pieces Δ' decreases strictly, so on a run between two pieces of the same sign, or between a piece and the end interval of the same sign, it keeps that sign. With the signs

at the ends, Δ' is therefore positive before the run of C pieces that follows the last P (or the left end) and negative after it, and on that run it decreases strictly from a positive to a negative value: it has exactly one zero, non-degenerate because $\Delta'' < 0$ there.

Then items (1)–(4) follow as in the proof of theorem 17. For δ_c: with t_c a numerical root and $h = 10^{-5}(r_0 - 2M)$, the enclosures give $\ell - 2R_0 > 0$ at $t_c - h$, $\ell - 2R_0 < 0$ at $t_c + h$, and $\Delta' < 0$ on $[t_c - h, t_c + h]$. Hence $\delta_c \in [\Delta(t_c + h), \Delta(t_c - h)]$, both enclosed. Uniqueness of the root t_* uses the single maximum, from theorem 17 for $q \geq 3/2$ and from the certificate for $q = 6/5$. □

The scripts, their logs and the list of all pieces with their labels and proved bounds are in the accompanying repository (`edges.py`, `unimodality.py`, `critical_separation.py`). The interior certificate was also run for the sixteen grid cases with $q \geq 3/2$, as an independent check of theorem 17; all pass.

B. The interior Maxwell angle

Computer-assisted part of the proof of Proposition X.8. We use the interval arithmetic of proposition X.6, with q an interval containing $1001/1000$, and set $\delta_{a,b} = \delta_i \mp 10^{-5}$ around a numerical tie δ_i. (A) On $N_1 = [21.33, 30]$ and $N_2 = [105, 140]$ (in units of M), $\mathrm{d}\Delta/\mathrm{d}t < 0$ in mean-value form, and $\Delta > \delta_b$ at the left end, $\Delta < \delta_a$ at the right end of each. So for each $\delta \in [\delta_a, \delta_b]$ there is exactly one dip $L_k(\delta)$ in N_k, and it minimizes F_δ on N_k. (B) The lengths $\ell_k(\delta)$ are enclosed at δ_a and δ_b; $G = \ell_2 - \ell_1$ satisfies $G(\delta_a) < 0 < G(\delta_b)$ and $G' = L_2 - L_1 > 0$ by lemma X.4, so there is exactly one tie δ_*. (C) No other candidate competes. Let U be the smaller upper bound of $\ell_1(\delta_b)$ and $\ell_2(\delta_b)$; since both lengths increase with δ, it suffices to show $F_\delta > U$ or $F_\delta > \ell_k(\delta)$ elsewhere. At $L = 0$, $2R_0 > U$. For $t \leq 2.15M$, $F_\delta \geq 2R_0 - (\pi - \delta_a)L > U$ by $\Delta < \pi$. For $t \geq 1300M$, $F_\delta \geq \delta_a L > U$ by $\mathcal{S} \geq 0$. In between, outside $N_1 \cup N_2$, cells in t are bisected until $F_\delta(L) - \ell_k(\delta) > 0$ for one k and all δ in the interval, using $\partial_\delta[F_\delta(L) - \ell_k(\delta)] = L - L_k(\delta)$: the test at δ_a suffices where $L \geq \max L_k$, the one at δ_b where $L \leq \min L_k$. The enclosure of Δ on a cell is intersected with $[0, \pi]$, which lemma X.3 justifies. The smallest certified margin is $3 \times 10^{-5} M$. The script and its log are in the accompanying repository (`maxwell_angle.py`). □

S9. THE EXAMPLES WITH A TIME-DEPENDENT FACTOR

This section contains the proofs, with all constants, of Theorems 23 and 24; the notation is that of section XI E of the main text.

A. The pulse at a small separation

Proof of Theorem 23. The proof follows that of Theorems 30 and 31. Without spin there is no magnetic term, the projective source reduces to ω_r/j with $j = |J|$, and the constants change; we list them. The localization and the fixed-arrival variational estimates use only $A \in C^1$ and $1/2 \leq A \leq 1$; the T-column additionally uses $T|(\log c_\varphi(A(T)))'| \leq 46/7$ on $[3/25, 21/100]$, verified below for the pulse and in Corollary XI.8 of the main text. No bound on A' along the whole trip is needed.

1. Localization. On the shell $\mathcal{S} = [23/4, 25/4]$ the radial density of $R_{1/2}$ is at least $1/(4\sqrt{2})$, so a curve leaving $\mathcal{S}$ and returning to $r = 6$ costs at least $1/(8\sqrt{2}) > 7/80$, while the circle $r = 6$ run with the actual clock costs at most $\rho_1(6)\delta = 3\sqrt{2}/50 < 7/80$. In $\mathcal{S}$ the angular cost per radian exceeds $4/5$ and $\omega > 11/100$, so every competitive curve lasts less than $35/44$. Since $2R_{1/2}(6) \geq \sqrt{2}$, Theorem 20 gives a global minimizer. At the arrival $\mathcal{H} = 0$, so $p_r(T) = c_r \cos\psi$, $J = c_\varphi \sin\psi$ with $c_r^2 = A^4/(2 - 2A^2/3) \in [3/88, 3/4]$ and $c_\varphi = \sqrt{24}\, c_r$, at $A = A(T)$. On the domain $r \in \mathcal{S}$, $|p_r| \leq 1$, $|J| \leq \sqrt{18}$, $1/2 \leq A \leq 1$ one

has $|\partial_r \mathcal{H}| < 3/20$. Since the radial end points coincide, p_r vanishes along the minimizer, whence $|p_r| < (3/20)T$, $J^2 > 4/9$, then $T < 1/5$, and, repeating the steps, $|p_r| < 3/100$, $J^2 > 64/81$, $|r-6| < 1/50$, $|\psi - s\pi/2| < 1/5$ with $s = \operatorname{sign} J$, and $T > 4/27$.

2. Uniqueness. For $(T,\theta) \in Q = [3/25, 21/100] \times [-1/5, 1/5]$ put $\psi = s\pi/2 + \theta$, integrate the canonical system backwards from the terminal state $(r(T), \varphi(T)) = (6, s/50)$ and set $F_s(T,\theta) = (-s\varphi(0), s(r(0)-6))$, whose zero means the launch $(6,0)$. The two values of s are the two reflected targets, not two minimizers of the same target. On Q, $j \geq (3/\sqrt{11})\cos(1/5) > 7/8$, every shot stays regular with $|r-6| < 3/40$ and $|p_r|/j < 79/1000$. At $\theta = 1/5$, with $\hat{p} = -sp_r$, the terminal value is $\hat{p}(T) = c_r \sin(1/5)$, and the force bound gives $\hat{p}(\eta) \geq \hat{p}(T) - (3/20)(21/100) > 0$, since $(3/88)(149/750)^2 > (63/2000)^2$ and $\sin x > x - x^3/6$; as $-s\dot{r} = a_H \hat{p}/S$, $F_2(T, 1/5) = \int_0^T (-s\dot{r})\,\mathrm{d}\eta > 0$, and symmetrically $F_2(T, -1/5) < 0$. Every minimizer is a zero of F_s in Q. On the returning shots $F_2 = 0$, the weight $jW = ja_H/S$ lies in $(350/153, 15/4)$, a ratio below $9/4$; the weighted mean bound with $c = 3/5$, as in the proof of Theorem 30, gives $|p_r(T)| \leq 189/10000$ and $|\theta| < 11/100$. With $\mathsf{a} = j\,\partial^2_{p_r}\mathcal{H}$, $\mathsf{b} = \partial_r \partial_{p_r}\mathcal{H}$, $\mathsf{l} = \partial^2_r \mathcal{H}/j$, $\mathsf{f} = \omega_r/j$, the coefficient bounds on Q are $7/4 < \mathsf{a} < 15/4$, $|\mathsf{b}| < 3/25$, $|\mathsf{l}| < 1/25$, $|\partial_r \partial_J \mathcal{H}| < 27/1000$, $\mathsf{f} < 1/20$, $|\partial_r \mathcal{H}|/j < 7/100$, and on the returning shots $\mathsf{a} > 2$, $|\mathsf{b}| < 11/200$, $|\partial_r \partial_J \mathcal{H}| < 23/1000$. For a variation put $u = \delta r$, $h = \delta J/J$, $P = (\delta p_r - (p_r/J)\delta J)/j$ and $Z = p_r/J$; the exact projective equations are

$$u' = \mathsf{b}u + \mathsf{a}P, \qquad P' = -\mathsf{l}u - \mathsf{b}P - \mathsf{f}h, \qquad \delta\varphi' = (\partial_r \partial_J \mathcal{H} + Z\,\partial_r \partial_{p_r}\mathcal{H})u - Zu',$$

with no magnetic source in the first one. For the T-column, with $\ell_T = [2/A(T) + A(T)/(3 - A(T)^2)]A'(T)$, the terminal data are $u(T) = -\dot{r}(T)$, $\delta\varphi(T) = -\dot{\varphi}(T)$, $h = \ell_T$ and $P(T) = \partial_r \mathcal{H}(T)/j$, the scale term $\ell_T p_r(T)$ canceling in $P(T)$. They give $\partial_\theta F_2 > (3/10)T$ on Q and, on $F_2 = 0$, $|\partial_\theta F_1|/\partial_\theta F_2 < 21/500$, $\partial_T F_1 > 9/100$, $|\partial_T F_2| < 21/50$, so $\det DF_s/\partial_\theta F_2 > 9/100 - (21/500)(21/50) > 7/100$. The T column uses $T|(\log c_\varphi(A(T)))'| \leq 46/7$. For the pulse $A' = -(\pi/2\tau)\sin(2\pi\eta/\tau)$, so $T|A'(T)| \leq \pi/2$ for $T \leq \tau$ and $A'(T) = 0$ for $T > \tau$; the factor $2/A + A/(3 - A^2)$ decreases on $[1/2, 1]$, with maximum $46/11$; and $(46/11)(\pi/2) < 46/7$ because $\pi < 22/7$. Lemma XII.10 gives at most one zero, so the minimizer is unique; its terminal state and covector determine the whole clocked curve.

3. The sector change. As in Corollary XII.11, $T \in C^1((0,\infty))$; step 1 gives $T > \tau$ for $\tau \leq 4/27$ and $T < \tau$ for $\tau \geq 1/5$. At a zero of $\Theta = T - \tau$, $A(T) = 1$, $j > 4$, $|p_r/J| \leq 63/8000$, $|\partial_r \mathcal{H}|/j < 3/80$, $|r-6| < 1/300$ and $|\partial_\theta F_1|/\partial_\theta F_2 < 13/1000$. Let $\theta^*(T,\tau)$ be the unique solution of $F_2 = 0$ and $\Gamma(T,\tau) = F_1(T, \theta^*(T,\tau); \tau)$. Along $T = \tau = \lambda$, in the clock $\nu = \eta/\lambda$ the terminal datum is fixed and $W = \partial_\lambda Y_\lambda$ solves $\mathrm{d}W/\mathrm{d}\nu = G + \lambda D_Y G\, W$, $W(1) = 0$; at the launch it equals the diagonal variation at fixed η, so $\Gamma_T + \Gamma_\tau = K_1 - (\partial_\theta F_1/\partial_\theta F_2)K_2$ with $K = (\partial_T + \partial_\tau)F_s$, as in Theorem 31. This gives $\Gamma_T + \Gamma_\tau > 49/500$, and $0 < \Gamma_T < 7/50$; hence $\Theta' < -7/10$ at every zero, which is therefore unique. Without spin the circular speed is $w = \sqrt{b_H} = \sqrt{1 - A^2 f/2}/\sqrt{h_\varphi}$, $h_\varphi = r^2/f$, and $0 \leq w - s\dot{\varphi} < 27/200000$. Put $\underline{r} = 1799/300$ and $\overline{r} = 1801/300$. On $[\underline{r}, \overline{r}]$, w decreases in r and is concave in A^2, and $\int_0^1 \alpha^2 = 19/32$ for $\alpha(\nu) = 1 - \frac{1}{2}\sin^2 \pi\nu$; hence

$$\frac{13}{24} w(\overline{r}, \tfrac{1}{2}) + \frac{11}{24} w(\overline{r}, 1) \leq \int_0^1 w(r(\nu), \alpha(\nu))\,\mathrm{d}\nu \leq \frac{\sqrt{1 - 19 f(\underline{r})/64}}{\sqrt{h_\varphi(\underline{r})}} < \frac{61}{500}.$$

The rational combination $(13/24)(1302/10^4) + (11/24)(1110/10^4)$ equals $607/5000$ exactly, and the left side exceeds it strictly because $w(\overline{r}, \frac{1}{2}) > 1302/10^4$ and $w(\overline{r}, 1) > 1110/10^4$. Hence $607/5000 - 27/200000 < \int_0^1 s\dot{\varphi}\,\mathrm{d}\nu < 61/500$. With $\tau^* \int_0^1 s\dot{\varphi}\,\mathrm{d}\nu = 1/50$ this gives the enclosure. The reflection $(\varphi, J) \mapsto (-\varphi, -J)$ preserves $\mathcal{H}$ and the radial data. □

B. The drop at a large separation

Proof of Theorem 24. (1) Frozen tail. Let V be the frozen value at $A = 1/2$ (so $q_0 = 8$) from (r, φ) to the target, on $\mathcal{D} = [599/100, 601/100] \times [-1/100, 1/100]$. Going radially to

$r = 6$ and then along the circle gives $V < 711/500 + 1/500 + 1/100 = 717/500$, using $\rho_{1/2}(6) = 3/\sqrt{11}$ and $3\pi/(2\sqrt{11}) < 711/500$, while $R_{1/2}(r) + R_{1/2}(6) > 359/250 - 1/500 = 717/500$ since $2R_{1/2}(6) > 359/250$. Here ${R'_{1/2}}^2 = r/(4(7r+2))$ and

$$R_{1/2}(r) = \frac{1}{98}\Big[7\big(\sqrt{r(7r+2)} - 4\sqrt{2}\big) + 2\sqrt{7}\,\operatorname{artanh}\frac{\sqrt{7}\,\big(4\sqrt{2} - \sqrt{r(7r+2)}\big)}{7r+16}\Big],$$

so the bounds on $R_{1/2}$ used here are evaluations of this closed form. Theorems 17 and 19 with reversibility select a unique frozen minimizer, non-degenerate by Corollary XI.6, so V is smooth on $\mathcal{D}$. Its length satisfies $1 < L < 3/2$, the lower bound by the Euclidean comparison with $143/200 < R_{1/2} < 81/100$ at both ends. A dip with periastron 4 sweeps more than $81/50 > \pi/2 + 1/100$: on $r \geq 4$ the curvature $K = 4(7r^3+14r^2+68r+8)/[r^3(r-2)^2(7r+2)]$ lies in $(0, 1/2]$, so $w = \mathrm{d}R/\mathrm{d}\rho$ increases, $w(4) > 9/10$, $\rho(4)/\rho(599/100) < 3/5 < \cos(9/10)$, and each half-angle $\int w\,\mathrm{d}\theta$ exceeds $81/100$. With the single maximum of the dip angle this puts the periastron of the selected geodesic above 4. Along it $0 < K \leq 1/2$, so the Jacobi field $j'' + Kj = 0$, $j(0) = 0$, $j'(0) = 1$, has $j \geq (13/16)s$ and $j' \geq 7/16$ for $s \leq 3/2$, and $0 \leq \operatorname{Hess} V < 16/13$. In the coordinates (r, φ) the frozen metric is $g_{rr} = r/(4(7r+2))$, $g_{\varphi\varphi} = r^2(r-2)/(4(7r+2))$; these are rational functions, and on $599/100 \leq r \leq 601/100$ they give $1/32 < g_{rr} < 1/25$, $g_{\varphi\varphi} < 1$, $|\Gamma^r_{rr}| < 1/100$, $|\Gamma^\varphi_{r\varphi}| < 1/4$ and $|\Gamma^r_{\varphi\varphi}| < 8$. With these bounds, $\|D^2 V\|_\infty < 7/2$; also $-V_\varphi > (7/5)(4/15)/(3/2) > 1/5$, using $\rho_{1/2}(4)^2 = 4/15$.

(1) Initial layer. Waiting at $r = 6$ for the clock h and then following the frozen circle costs less than $711/500 + h/2 < 2R_{1/2}(6)$, so Theorem 20 gives a minimizer. Since $|\dot r| < 1$ and $|\dot\varphi| \leq 1/(3\sqrt{3}) < 1/5$, every minimizer arrives after h and has $x(h) \in \mathcal{D}$; by the principle of optimality its tail is the frozen minimizer from $x(h)$, so $p(h) = -DV(x(h))$. On $r \in [5, 7]$, $|p_r| \leq 1$, $1/10 \leq J \leq 2$, $1/2 \leq A \leq 1$ the canonical field G satisfies $\|G\|_\infty < 1$ and $\|D_Y G\|_\infty < 300$, uniformly in the value of A and without A'. Indeed, with $d = 1 - A^2 f/2$, $k_1 = f^2 d$, $k_2 = fd/r^2$ and $S^2 = k_1 p_r^2 + k_2 J^2$, there $3/5 \leq f \leq 5/7$, $9/14 \leq d \leq 1$, $k_1 \leq 25/49$, $1/128 < k_2 \leq 1/25$ and $1/120 < S < 1$, while $|(\log k_i)'| < 3/5$ and $|(\log k_i)''| < 1$ in r. Averaging with the weights $k_1 p_r^2/S^2$ and $k_2 J^2/S^2$ gives $|S_r| < 3/10$ and $|S_{rr}| < 77/100$, hence $|H_r|, |H_{rr}| < 1$; also $|H_{p_i}| \leq \sqrt{k_i}$ and $|H_{rp_i}| \leq (3/5 + 3/10)\sqrt{k_i} < 1$. Finally $D^2_{pp} S = S^{-1}\Lambda(I - \xi\xi^\top)\Lambda$ with $\Lambda = \operatorname{diag}(\sqrt{k_1}, \sqrt{k_2})$ and $\xi = \Lambda p/S$, so its entries are at most $\max(k_1, k_2)/S < 120 \cdot 25/49$; the column of φ vanishes, and the row sums of $D_Y G$ are below $1 + 2 \cdot 120 \cdot 25/49 < 124$. Let $E_h(x)$ be the state reached at the clock 0 by the backward flow from $(x, -DV(x))$. By Grönwall, $\|DE_h - I\|_\infty \leq 4(e^{300h} - 1) \leq 12/97 < 1$, so E_h is injective on the convex set $\mathcal{D}$. All minimizers satisfy $E_h(x(h)) = (6, 0)$, hence share the state and covector at h, and the regular flow fixes the whole curve. For the drop, $h = d + \epsilon \leq 10^{-4}$.

(2) Late drops. Suppose a minimizer exists with cost $C \leq B_d$. Since $A \leq 1$ and the density of $R_{1/2}$ is at most a quarter of that of R_1, $B_d \leq R_1(6) + R_{1/2}(6) \leq \frac{5}{4} R_1(6) \leq \frac{5}{2}\sqrt{3} < \frac{13}{3}$. A curve arriving by d costs at least $\sqrt{2} R_1(6) > B_d$, by the Euclidean comparison, so the minimizer is still traveling at d. With $s_0 = \min_{[0,d]} r$, the plateau costs at least $df(s_0)/\sqrt{2}$, so $s_0 - 2 < 13/(d - 13/2)$. If the periastron of Lemma XI.10 precedes d, comparing the radial barriers before and after it with $C \leq B_d$ gives $\frac{3}{4} R_1(r(d)) \leq 2R_1(s_0)$, hence $r(d) - 2 < 4(s_0 - 2)$; otherwise $r(d) = s_0$. Thus $r(d) - 2 < 52/(d - 13/2) < 10^{-5}$. On the plateau p_η is constant and $S = \omega_1 - p_\eta$, so $0 \leq p_\eta < \omega_1(r(d)) < 1/200000$ and $J \leq \rho_1(r(d)) < 1/300$. On the plateau the orbit is that of the frozen metric with $\rho_{\mathrm{pl}} = \rho_1(1 - p_\eta/\omega_1)$, $R'_{\mathrm{pl}} = R'_1(1 - p_\eta/\omega_1)$, and $\mathrm{d}\rho_{\mathrm{pl}}/\mathrm{d}R_{\mathrm{pl}} - \mathrm{d}\rho_1/\mathrm{d}R_1 = \rho_1 p_\eta \omega_1'/[\omega_1^2 R'_1(1 - p_\eta/\omega_1)] \geq 0$, so $w_{\mathrm{pl}} \leq w_1 \leq r\sqrt{f}/(r-1)$; after the drop $p_\eta = 0$ and $w_{1/2} \leq r\sqrt{f}/(r-1)$ as well. Split each frozen part at $r_s = 201/100$: inside, $w < 1/7$ and each leg sweeps at most $\pi/14$; outside, $w < 1$, $\rho > 1/40$ and each leg sweeps at most $\arcsin(40J) < 7/50$. Each frozen part has at most two legs on each side of r_s, and the drop lasts ϵ with $|\dot\varphi| < 1/5$. Hence the total angle is less than $2\pi/7 + 14/25 + 1/500 < \pi/2$, a contradiction. So no minimizer exists, and $m_d = B_d$ by Theorem 21. The argument covers a minimizer of cost B_d.

(3) First contact. Fix D_0. For $d \leq D_0$ every minimizer has $r \leq 6$, and $r - 2 \geq 4e^{-(D_0+\epsilon)/2}$ up to the clock $D_0 + \epsilon$. If $J \to 0$ along minimizers, then for small J one has $p'_r > 1/400$

in the zone $|p_r| \le \sqrt{J}$, which is therefore crossed once, in a clock at most $800\sqrt{J}$, while $\dot\varphi \le \sqrt{J}/4$ outside it. If $T \le h = d + \epsilon$ there is no frozen tail; if $T > h$, since $r(h) \le 6$ and $\rho_{1/2}$ is strictly increasing, the frozen tail, extended backwards to $r = 6$ through its turning point if necessary, is a subarc of the equal-radius dip with Clairaut constant J; its angle is at most $\Delta_{1/2}(J; 6)$, which tends to zero as $J \to 0$ (Lemma X.3). The total angle would tend to zero, so $J \ge J_*(D_0) > 0$: the minimizers stay away from $r = 2$, have bounded duration and form a compact family. The price B_d is continuous in d. The value m_d is continuous: for $d < d_0$, $A_d \le A_{d_0}$ pointwise, so every relaxed competitor at d_0 is admissible at d with no larger cost; for $d > d_0$, waiting $d - d_0$ at the launch and then following a relaxed competitor at d_0 gives $m_d \le m_{d_0} + (d - d_0)\omega_1(6)$. These comparisons are made in the relaxed class; Lemma XI.1 identifies its infimum with the rail infimum. Finally, along $d_n \to d_0$, either $m_{d_n} = B_{d_n} \to B_{d_0} \ge m_{d_0}$, or compactness gives a limit competitor of cost $\liminf m_{d_n} \ge m_{d_0}$. The set $\{m_d = B_d\}$ is therefore closed, not empty by (2), and disjoint from $[0, 1/20000]$, where $m_d < 2R_{1/2}(6) \le B_d$. For $d < d_*$ Theorem 21 gives existence, and the limit of minimizers as $d \uparrow d_*$ is a minimizer of cost B_{d_*}. As no minimizer exists at 10^7, $d_* < 10^7$. □

S10. THE EXAMPLE ON THE EQUATOR WITH SPIN

This section contains the proofs, with all constants, of Proposition XII.8 and Theorems 30 and 31; the data and notation are those of the corresponding subsection of the main text.

A. Localization

Proof of Proposition XII.8. Competitive curves. By Lemma XII.1(1), $C \ge L_{1/2}$. On $\mathcal{S}$ the radial coefficient of $L_{1/2}$ is at least $(4\sqrt{2})^{-1}\sqrt{345/349}$, since $N \ge 1$ and $r(r-2)/\Delta \ge 345/349$. A curve from and to $r = 6$ that reaches $r = 6 \pm 1/4$ crosses that half-shell twice, so $C \ge (8\sqrt{2})^{-1}\sqrt{345/349} > 7/80$. The angular coefficient of $L_{1/2}$ is at least $\sqrt{349}/(16\sqrt{2}) > 4/5$, so $\operatorname{Var}\varphi < (5/4)(7/80) = 7/64$, which excludes every other lift. By Lemma XII.1(1), $c_A \ge \omega/(1 + |\beta|_{h_0}) > (11/25)A^2 \ge 11/100$ on $\mathcal{S}$, hence $T < 35/44$. The circle $r = 6$ run with the actual clock costs at most its frozen length at $A = 1$, $\sqrt{291}/200$.

Force. On $\mathcal{D} = \{r \in \mathcal{S},\ |p_r| \le 1,\ |J| \le \sqrt{291}/4,\ A \in [1/2, 1]\}$ every element of the Clarke differential of $\mathcal{H}$ in r has modulus at most $744633/5000000 < 3/20$. Indeed, with $u = \sqrt{a_H}\, p_r$, $w = \sqrt{b_H}\, \xi$, $\alpha = \frac{1}{2}\partial_r \log a_H$ and $\gamma = \frac{1}{2}\partial_r \log b_H$,

$$\partial_r S = \frac{\alpha u^2 + \gamma w^2}{S} + \frac{\sqrt{b_H}\, \xi_r\, w}{S}, \qquad \Big|\frac{\alpha u^2 + \gamma w^2}{S}\Big| \le \max\{\alpha|u|, |\gamma||w|\},$$

because $0 \le \alpha \le 11/100$ and $-77/500 \le \gamma \le 0$ on the shell. More precisely,

$$\max\{\alpha|u|, |\gamma||w|\} \le \max\left\{\frac{11}{100}\frac{2}{3}, \frac{77}{500}\frac{67}{500}\frac{22}{5}\right\} = \frac{77}{500}\frac{67}{500}\frac{22}{5}.$$

With $\sqrt{a_H} < 2/3$, $\sqrt{b_H} < 67/500$, $|\xi| < 22/5$, $\omega_r < 43/1000$, $\xi_r = g/r < 43/2000$, $|k_H| < 6/1250$ and $|\partial_r k_H| < 69/25000$, this gives

$$|\partial_r \mathcal{H}| \le \frac{43}{1000} + \frac{77}{500}\frac{67}{500}\frac{22}{5} + \frac{67}{500}\frac{43}{2000} + \frac{69}{25000}\frac{22}{5} + \frac{6}{1250}\frac{43}{2000} = \frac{744633}{5000000} < \frac{3}{20}.$$

Here, with $B_m = |\beta|^2_{h_0}$,

$$\ell_r = \frac{2}{r} - \frac{2}{r(r-2)} - \frac{2r-2}{\Delta}, \qquad \ell_\varphi = \frac{2r-2}{\Delta} - \frac{4}{r(r-2)},$$
$$L_N = \frac{N_r}{N} = \frac{NA^2}{r^2}, \qquad L_d = \frac{d_r}{d}.$$

On $\mathcal{S}$ direct rational comparisons give

$$0 \le L_N \le \frac{800}{17457}, \quad 0 \le L_d < \frac{47}{1000}, \quad 2L_d - L_N < \frac{6}{125},$$
$$-\frac{11}{50} < \ell_r < -\frac{1}{10}, \quad \frac{1}{5} < \ell_\varphi < \frac{13}{50}.$$

For example, setting $B_* = 256/(23^2 \cdot 349)$, $N_* = 184/169$ and $L_* = 800/17457$, the identity $(\log B_m)_r = -2/r - \Delta'/\Delta$ gives

$$0 \le L_d - L_N \le \frac{B_*(4/5 + L_*)}{N_* - B_*}.$$

The displayed bounds imply $\alpha = -(L_d + \ell_r)/2 \in [0, 11/100]$ and $\gamma = (L_N - 2L_d - \ell_\varphi)/2 \in [-77/500, 0]$. Also $\partial_r k_H = -k_H[1/(r-2) + L_d + \ell_\varphi]$. The bounds on a_H, b_H, k_H follow from $d \ge N_* - B_*$ and the shell bounds above; for b_H, the function $N/(N - B_*)^2$ decreases in N. The same force bound holds for the convex hull of limiting gradients at $S = 0$.

Bootstrap. A minimizer is a normal extremal (Proposition XII.2) with J constant and, by transversality, $p_r(T) = c_r \cos\psi$, $J = c_\varphi \sin\psi$ at $A = A(T)$. Up to a first exit from $\mathcal{D}$ one has $|p_r| \le \sqrt{72/97} + (3/20)(35/44) < 1$, so no exit occurs. Where $p_r \ne 0$ the support is regular and $\dot r$ has the sign of p_r; since $r(0) = r(T) = 6$, p_r vanishes somewhere and $|p_r| \le (3/20)T < 21/176$. The terminal ellipse then gives $J^2 > 4/9$, and $|g| < 1/8$ gives $|\xi| > 13/24$: the support is regular along the whole curve. With $w_* = |p_r|\sqrt{h_\varphi/h_r}/|\xi|$,

$$\operatorname{sign}(\xi)\,\dot\varphi = \frac{\sqrt{N}}{d\sqrt{h_\varphi}\sqrt{1 + (1 - |\beta|^2/N)w_*^2}} - \operatorname{sign}(\xi) k_H \ge \frac{a_0}{\sqrt{1 + BT^2}} - d_0,$$

$a_0^2 = 33/2900$, $B = 9396/4225$, $d_0 = 6/1250$. The right-hand side is positive for $T \le 35/44$, so φ is monotone and $\operatorname{sign} J = \operatorname{sign}\xi = s$; the function $T(a_0/\sqrt{1+BT^2} - d_0)$ increases on $[0, 35/44]$ and exceeds $1/50$ at $T = 21/100$, hence $T < 21/100$. Repeating the steps, $|p_r| < (3/20)(21/100) = 63/2000$, $J^2 > 64/81$, $|\xi| > 55/72$ and $|p + \omega\beta|_{h_0^{-1}} \ge |\xi|/\sqrt{h_\varphi} > 55/(72\sqrt{58}) > 1/10$. Since $\dot r = p_r/(h_r\sqrt{N\xi^2/h_\varphi + dp_r^2/h_r})$, $|\dot r| < (63/2000)(36\sqrt{58}/55)$ and $|r-6| \le (T/2)\sup|\dot r| < 1/50$. Finally $|\cos\psi(T)| = |p_r(T)|/c_r < 63\sqrt{1067}/12000 < \sin(1/5)$ with $\sin\psi(T)$ of sign s, and $|\dot\varphi| \le 1/(\sqrt{h_\varphi} - |\beta_\varphi|) < 1/6$ gives $T > 3/25$. □

B. Uniqueness

Proof of Theorem 30. Shooting map. For $(T, \theta) \in Q_s = [3/25, 21/100] \times [-1/5, 1/5]$ put $\psi = s\pi/2 + \theta$, so that $p_r(T) = -sc_r \sin\theta$ and $J = sc_\varphi \cos\theta$ at $A = A_\tau(T)$. Integrate the canonical system backwards from $(x_f, p(T))$ and set $F_s(T, \theta) = (-s\varphi(0), s(r(0) - 6))$. Since J is constant, $j \ge c_\varphi(1/2)\cos(1/5) > 8/9$. Up to a first exit from $r \in \mathcal{S}$, $|p_r| \le 1$, one has $|\xi| > j - 1/8 > 55/72$, so the support is regular. The rail speed bound $|\dot r| < 1$ gives $|r - 6| < T \le 21/100 < 1/4$, while the force bound gives $|p_r| \le \sqrt{72/97}/5 + 63/2000 < 1$. Neither boundary can be reached; the regular field therefore extends to $\eta = 0$ for every point of Q_s. Moreover,

$$\frac{|p_r|}{j} \le \kappa_6 |\tan\theta| + \frac{63}{2000j} < \frac{203}{1000}\frac{10}{49} + \frac{567}{16000} < \frac{77}{1000}, \qquad \frac{55}{64} < \frac{|\xi|}{j} < \frac{73}{64}.$$

Since $a_H/\sqrt{b_H} = \sqrt{h_\varphi}/(h_r\sqrt{N}) < 15/4$, $|\dot r| < 84/250$ and $|r - 6| < 3/40$ along every shot. Thus the whole family stays in $\mathcal{D}$, is regular, and $F_s \in C^1(Q_s)$, because $A_\tau \in C^1$. By Proposition XII.8 every minimizer is a zero of F_s with $|\theta| < 1/5$, and a minimizer exists (Remark XII.7). It suffices to verify the hypotheses of Lemma XII.10. Write $D_{iT} = \partial_T F_i$, $D_{i\theta} = \partial_\theta F_i$.

Signs on the sides. Put $\hat{p} = -sp_r$. At $\theta = 1/5$ the terminal value is $\hat{p}(T) = c_r \sin(1/5) \geq (6/\sqrt{1067})(149/750)$, and the force bound gives $\hat{p}(\eta) \geq \hat{p}(T) - (3/20)(21/100) > 0$, since $(36/1067)(149/750)^2 > (63/2000)^2$. As $-s\dot{r} = a_H\hat{p}/S$, we get $F_2(T, 1/5) = \int_0^T (-s\dot{r})\,\mathrm{d}\eta > 0$; symmetrically $F_2(T, -1/5) < 0$.

Returning shots. If $F_2 = 0$, then $r(0) = r(T) = 6$ and p_r vanishes somewhere. The force bound gives $|p_r| \leq 63/2000$; the radial speed estimate and the two equal radial end points give $|r - 6| < 1/50$ along the shot. Put $w = a_H/S$ and $t_H = a_H p_r^2/(b_H\xi^2)$. Then

$$\int_0^T w p_r \,\mathrm{d}\eta = 0, \qquad p_r(\eta) = p_r(T) + \int_\eta^T \partial_r \mathcal{H}(v)\,\mathrm{d}v.$$

On the shell, $h_r < 12/5$, $h_\varphi > 49$, $N < 25/16$, hence $7/3 < a_H/\sqrt{b_H} < 15/4$. Also $a_H/b_H < 29$, so $t_H < 1/20$ and $\sqrt{1+t_H} < 41/40$. Consequently

$$m := \frac{7}{3}\frac{64}{73}\frac{40}{41} = \frac{17920}{8979} < jw = \frac{a_H/\sqrt{b_H}}{(|\xi|/j)\sqrt{1+t_H}} < \frac{15}{4}\frac{64}{55} = \frac{48}{11} =: M,$$

and $M/m = 26937/12320 < 9/4$. For any weight $\mathcal{W}$ with $m \leq \mathcal{W} \leq M$ on $[0,1]$, and $c = 3/5$,

$$\int_0^1 (x-c)\mathcal{W}(x)\,\mathrm{d}x \leq M\int_c^1 (x-c)\,\mathrm{d}x + m\int_0^c (x-c)\,\mathrm{d}x = \frac{M(1-c)^2 - mc^2}{2} \leq 0.$$

Apply this with $x = (T-\eta)/T$ and $\mathcal{W}(x) = jw(T(1-x))$. It gives the weighted mean bound

$$|p_r(T)| \leq \frac{3}{20}\frac{\int_0^T w(T-\eta)\,\mathrm{d}\eta}{\int_0^T w\,\mathrm{d}\eta} \leq \frac{3}{20}\frac{3T}{5} \leq \frac{189}{10000}.$$

Since $c_r \geq 6/\sqrt{1067}$,

$$\frac{36}{1067}\left[\frac{11}{100} - \frac{1}{6}\left(\frac{11}{100}\right)^3\right]^2 > \left(\frac{189}{10000}\right)^2$$

and $\sin x > x - x^3/6$ on $(0, 1/5]$ imply $|\theta| < 11/100$.

Coefficients. Let $Z = p_r/\xi$, $\epsilon = g/j < 9/64$, and

$$\mathsf{a} = j\,\partial_{p_r}^2\mathcal{H}, \quad \mathsf{b} = \partial_r\partial_{p_r}\mathcal{H}, \quad \mathsf{l} = \partial_r^2\mathcal{H}/j, \quad \mathsf{f} = \big(\omega_r - \xi_r\dot{\varphi} + g\,\partial_r\partial_J\mathcal{H}\big)/j, \quad \mathsf{q} = \mathsf{a}Z\epsilon.$$

On the whole family, and (in parentheses) on returning shots,

$$\begin{aligned} &\tfrac{29}{20}\,(\tfrac{37}{20}) < \mathsf{a} < \tfrac{48}{11}, \quad |\mathsf{b}| < \tfrac{7}{50}\,(\tfrac{8}{125}), \quad |\mathsf{l}| < \tfrac{1}{25}, \\ &|\partial_r\partial_J\mathcal{H}| < \tfrac{4}{125}\,(\tfrac{13}{500}), \quad |\mathsf{f}| < \tfrac{57}{1000}\,(\tfrac{7}{125}), \quad |\mathsf{q}| < \tfrac{11}{200}\,(\tfrac{13}{500}), \end{aligned}$$

and $|\partial_r\mathcal{H}|/j < 2/25$. These follow from the exact identities, with $\rho = a_H p_r^2/S^2$ and $\sigma = 1 - \rho$,

$$\mathsf{a} = \frac{a_H/\sqrt{b_H}}{(|\xi|/j)(1 + a_H p_r^2/(b_H\xi^2))^{3/2}}, \qquad \mathsf{b} = \dot{r}\big[2\alpha - \alpha\rho - \gamma\sigma - (\xi_r/\xi)\sigma\big],$$

$$\partial_r\partial_J\mathcal{H} = (\dot{\varphi} + k_H)\big[\gamma + \rho(\gamma - \alpha + \xi_r/\xi)\big] - \partial_r k_H,$$

$$\frac{\partial_r^2 S}{S} = (\alpha' + \alpha^2)\rho + (\gamma' + \gamma^2)\sigma + \rho\sigma\Big(\gamma - \alpha + \frac{\xi_r}{\xi}\Big)^2 + \frac{b_H\xi(2\gamma\xi_r + \xi_{rr})}{S^2},$$

$\partial_r^2\mathcal{H} = -\omega_{rr} + \partial_r^2 S - \partial_r^2 k_H\,\xi - 2\partial_r k_H\,\xi_r - k_H\xi_{rr}$, together with $|\alpha'| < 8/125$, $|\gamma'| < 57/1000$, $|\partial_r^2 k_H| < (6/1250)[(23/40)^2 + 3/20]$, $|\omega_{rr}| < 3/200$, $|\xi_{rr}| < 3/400$ on $\mathcal{S}$. The last group comes from $\alpha = -(L_d + \ell_r)/2$, $\gamma = (L_N - 2L_d - \ell_\varphi)/2$, $\partial_r k_H = -k_H C_k$ with $C_k = 1/(r-2) + L_d + \ell_\varphi$, where $\ell_r = \partial_r \log h_r$, $\ell_\varphi = \partial_r \log h_\varphi$, $L_N = N_r/N = NA^2/r^2$, $L_d = d_r/d$, and from $\ell_r' = -2/r^2 + P - 2/\Delta + V^2$, $\ell_\varphi' = 2/\Delta - V^2 + 2P$, $L_N' = L_N(L_N - 2/r)$, $L_d' = (N'' - (|\beta|^2)'')/d - L_d^2$, with $V = \Delta'/\Delta$ and $P = 4(r-1)/(r^2(r-2)^2)$ both decreasing on $\mathcal{S}$; every term is evaluated at the appropriate end of the shell. For completeness, in these coefficient identities primes denote r-derivatives. Writing $B_m = |\beta|^2_{h_0}$, the endpoint comparisons give

$$0 < \ell_r' < \frac{53}{500}, \qquad |\ell_\varphi'| < \frac{7}{125}, \qquad |L_N'| < \frac{2}{125}, \qquad |L_d'| < \frac{21}{1000}.$$

Indeed, $L_N' = L_N(L_N - 2/r)$, $N/d < 501/500$ and $N''/N = 2L_N^2 - 2L_N/r$. With $L_B = (\log B_m)' = -2/r - \Delta'/\Delta$, one has $|L_B| < 4/5$, $|L_B'| < 13/50$ and $|B_m''| < (9/10)B_m$. Thus, with $r_- = 23/4$, $B_* = 256/(23^2 \cdot 349)$, $d_* = 184/169 - B_*$,

$$|L_d'| \le \frac{501}{500}\frac{800}{17457}\frac{2}{r_-} + \frac{9}{10}\frac{B_*}{d_*} + \left(\frac{47}{1000}\right)^2 < \frac{21}{1000}.$$

The bounds on ℓ_r', ℓ_φ' follow by evaluating each term in the displayed formulas at the two shell endpoints. These yield the stated bounds on α', γ' and on $k_{H,rr}$. Put $t_H = a_H p_r^2/(b_H \xi^2)$. On Q_s, $t_H < 6/25$ and $\rho < 6/31$; on returning shots $t_H < 1/20$ and $\rho < 1/21$. Also

$$|\xi_r/\xi| < \frac{43}{2000}\frac{72}{55}, \quad |\gamma - \alpha + \xi_r/\xi| < \frac{3}{10},$$
$$|\alpha'| + \alpha^2 < \frac{81}{1000}, \quad |\gamma'| + \gamma^2 < \frac{81}{1000}, \quad S/j < \frac{43}{250}.$$

In particular, setting $K_2 = (6/1250)[(23/40)^2 + 3/20]$, the second-derivative identity gives the explicit comparison

$$\begin{aligned} |\mathcal{H}_{rr}|/j &< \frac{43}{250}\left(\frac{81}{1000} + \frac{6}{31}\frac{9}{100}\right) + \frac{67}{500}\left(2\frac{77}{500}\frac{43}{2000} + \frac{3}{400}\right)\frac{9}{8} \\ &\quad + \frac{3}{200}\frac{9}{8} + K_2\frac{73}{64} + 2\frac{69}{25000}\frac{43}{2000}\frac{9}{8} + \frac{6}{1250}\frac{3}{400}\frac{9}{8} \\ &= \frac{9606889437}{248000000000} < \frac{1}{25}. \end{aligned}$$

For b, the bracket in its identity has modulus below $11/50 + 77/500 + (43/2000)(72/55)$; multiply by $84/250$ on Q_s, or by $(15/4)(63/2000)(72/55)$ on returning shots. For $\mathcal{H}_{rJ}$ use $|\dot\varphi + k_H| < 67/500$ and the two bounds on ρ above. For a use $7/3 < a_H/\sqrt{b_H} < 15/4$ and $(1+t_H)^{3/2} < 7/5$ on Q_s, or $< 11/10$ on returning shots. The remaining bounds follow from

$$|\mathsf{f}| \le \frac{\omega_r}{j} + \frac{\xi_r}{j}|\dot\varphi| + \epsilon|\mathcal{H}_{rJ}|, \qquad |\mathsf{q}| \le \frac{48}{11}|Z|\frac{9}{64}, \qquad |\dot\varphi| < \frac{7}{50}.$$

Finally, preserving the maximum in the force estimate,

$$|\mathcal{H}_r|/j < \frac{43}{1000}\frac{9}{8} + \frac{77}{500}\frac{67}{500}\frac{73}{64} + \frac{67}{500}\frac{43}{2000}\frac{9}{8} + \frac{69}{25000}\frac{73}{64} + \frac{6}{1250}\frac{43}{2000}\frac{9}{8} < \frac{2}{25}.$$

Variations. For a variation of a shot let $u = \delta r$, $h = \delta J/J$ (constant) and $P = (\delta p_r - (p_r/J)\delta J)/j$. Differentiating the canonical equations gives exactly

$$u' = \mathsf{b}u + \mathsf{a}P - \mathsf{q}h, \qquad P' = -\mathsf{l}u - \mathsf{b}P - \mathsf{f}h, \qquad \delta\varphi' = \big(\partial_r\partial_J\mathcal{H} + Z\,\partial_r\partial_{p_r}\mathcal{H}\big)u - Zu',$$

the last because the Hessian of $\mathcal{H}$ in p has rank one: $\partial_{p_r}\partial_J\mathcal{H} = -Z\partial_{p_r}^2\mathcal{H}$ and $\partial_J^2\mathcal{H} = -Z\partial_{p_r}\partial_J\mathcal{H}$.

The θ-column on Q_s. Here $u(T)=0$, $h=-\tan\theta$, $|h|\le 10/49$ and $P(T)=-s\kappa_6/\cos^2\theta$. In the reversed time $t=T-\eta$, $U=su$ and $V=-sP$ satisfy $\mathrm{d}U/\mathrm{d}t=-\mathsf{b}U+\mathsf{a}V+s\mathsf{q}h$ and $\mathrm{d}V/\mathrm{d}t=-\mathsf{l}U+\mathsf{b}V-s\mathsf{f}h$. Set $T_*=21/100$ and, for $B\in\{7/50,8/125\}$,

$$\mathcal{M}_B=\begin{pmatrix}1-T_*B & -T_*48/11\\ -T_*/25 & 1-T_*B\end{pmatrix}.$$

Its diagonal and determinant are positive, and $\mathcal{M}_B^{-1}$ is entrywise nonnegative. With $U_\infty=\sup|U|$, $V_\infty=\sup|V|$, integration of the two equations gives the componentwise inequality

$$\mathcal{M}_{7/50}\begin{pmatrix}U_\infty\\ V_\infty\end{pmatrix}\le\begin{pmatrix}T_*(11/200)(10/49)\\ (203/1000)/(49/50)^2+T_*(57/1000)(10/49)\end{pmatrix}.$$

Multiplication by the nonnegative inverse gives $U_\infty<213/1000$, $V_\infty<223/1000$. Since $V(0)=\kappa_6/\cos^2\theta>1/5$, $V>1/5-(21/100)[(1/25)(213/1000)+(7/50)(223/1000)+(57/1000)(10/49)]>189/1000$ and $\mathrm{d}U/\mathrm{d}t>-(7/50)(213/1000)+(29/20)(189/1000)-(11/200)(10/49)>23/100$. Thus U increases from 0, and

$$D_{2\theta}=U(T)>\tfrac{23}{100}T\ge\tfrac{69}{2500}>\tfrac{1}{40}\quad\text{on } Q_s.$$

On a returning shot, since $\delta\varphi(T)=0$, $\int_0^T|u'|=D_{2\theta}$ and $\int_0^T|u|\le TD_{2\theta}$, the angular identity gives $|D_{1\theta}|\le(Z_0+C_0T)D_{2\theta}$ with $Z_0=(63/2000)(72/55)$ and $C_0=13/500+(8/125)Z_0$; hence $|D_{1\theta}|/D_{2\theta}<6/125$.

The T-column on returning shots. With $\ell_T=c_\varphi'(T)/c_\varphi=[2/A+A/(3-A^2)]A'(T)$, which also equals c_r'/c_r, the moving end point gives $u(T)=-\dot r(T)$, $\delta\varphi(T)=-\dot\varphi(T)$, $h=\ell_T$ and $P(T)=\partial_r\mathcal{H}(T)/j$: the term $\ell_Tp_r(T)$ cancels. For the pulse $T|\ell_T|\le(46/11)(\pi/2)<46/7$, since $A'(T)=0$ for $T\ge\tau$ and $T|A'(T)|\le\pi/2$ otherwise. Here $|u(T)|<93/1000$, $|P(T)|<2/25$. With $U_T=\sup|u|$, $V_T=\sup|P|$ the integral comparison is

$$\mathcal{M}_{8/125}\begin{pmatrix}U_T\\ V_T\end{pmatrix}\le\begin{pmatrix}93/1000+(46/7)(13/500)\\ 2/25+(46/7)(7/125)\end{pmatrix}.$$

It gives

$$U_T\le\frac{4035532775}{5808706624}<\frac{7}{10},\qquad V_T\le\frac{1526921275}{3319260928}<\frac{47}{100}.$$

Using these fractions before rounding, the first variational equation gives

$$\int_0^T|u'|\,\mathrm{d}\eta\le T_*\left[\frac{8}{125}U_T+\frac{48}{11}V_T\right]+\frac{46}{7}\frac{13}{500}\le\frac{436915382371}{726088328000}<\frac{61}{100}.$$

Hence $|D_{2T}|<7/10$. On the returning domain $|r-6|<1/50$, $N\le602/401$, $h_\varphi\le2769038363/50501250$ imply $b_H>1/(Nh_\varphi)>(11/100)^2$. Using $D_{1T}=s\dot\varphi(T)+s\int_0^T\delta\varphi'\,\mathrm{d}\eta$ and $s\dot\varphi(T)\ge\sqrt{b_H}/\sqrt{1+1/20}-|k_H|$, we obtain

$$D_{1T}>\tfrac{11}{100}\cdot\tfrac{20}{21}-\tfrac{6}{1250}-C_0\cdot\tfrac{21}{100}\cdot\tfrac{7}{10}-Z_0\cdot\tfrac{61}{100}>\tfrac{7}{100}.$$

Conclusion. On $\{F_2=0\}$, $\det DF_s/D_{2\theta}=D_{1T}-(D_{1\theta}/D_{2\theta})D_{2T}>7/100-(6/125)(7/10)=91/2500>0$. Lemma XII.10 leaves at most one zero of F_s, hence exactly one minimizer datum (T,θ). It fixes the terminal state and normalized covector; the regular canonical field and the non-degenerate support then fix the whole timed curve. □

C. The sector change

Proof of Theorem 31. Regularity. The proof of Corollary XII.11 already gives $T_s \in C^1((0,\infty))$. By Proposition XII.9, every zero of $\Theta_s(\tau) = T_s(\tau) - \tau$ lies in $(\tau_{\mathrm{no}}, 1/5)$, and there is at least one.

At a zero. There $T = \tau < 1/5$ and $A(T) = 1$, so $j = (\sqrt{291}/4)\cos\theta > 4$, $|\xi| > 31/8$ and, on the returning shot, $|p_r/\xi| \le Z_* = 63/7750$, $a_H p_r^2/(b_H\xi^2) < 1/500$, $|\partial_r \mathcal{H}|/j < 3/80$, $|\dot r| < 31/1000$, $|\dot r(T)| < 19/1000$ and $|r-6| < 1/300$. As in the proof of Theorem 30, $|D_{1\theta}|/D_{2\theta} < Z_* + (21/100)C_* < 7/500$ with $C_* = 13/500 + (8/125)Z_*$.

Diagonal derivative by dilation. Fix θ and, for λ near τ, shoot with $T = \tau = \lambda$. In the clock $\nu = \eta/\lambda$ the profile is $\alpha(\nu)$ for every λ, and the terminal datum does not depend on λ, because $A_\lambda(\lambda) = 1$. The backward state $Y_\lambda(\nu)$ solves $\mathrm{d}Y_\lambda/\mathrm{d}\nu = \lambda G(\alpha(\nu), Y_\lambda)$, with G the canonical field, so $W = \partial_\lambda Y_\lambda$ solves $\mathrm{d}W/\mathrm{d}\nu = G + \lambda D_Y G\, W$ with $W(1) = 0$ and $\delta J = 0$. In the clock η, with $u_d = W_r$, $P_d = W_{p_r}/j$ and $w_d = W_\varphi$,

$$u_d' = \mathsf{b} u_d + \mathsf{a} P_d + \frac{\dot r}{\tau}, \qquad P_d' = -\mathsf{l} u_d - \mathsf{b} P_d - \frac{\partial_r \mathcal{H}}{\tau j}, \qquad w_d' = \partial_r \partial_J \mathcal{H}\, u_d - \mathsf{a} Z P_d + \frac{\dot\varphi}{\tau},$$

with zero data at $\eta = T$. Here W is the variation at fixed ν; the variation along the diagonal at fixed η is $W - (\eta/\tau)G$, evaluated at $\nu = \eta/\tau$. They agree at $\eta = 0$, so $K = (\partial_T + \partial_\tau)F_s = (-s w_d(0), s u_d(0))$. Put $U_d = \sup|u_d|$, $V_d = \sup|P_d|$. Since $T/\tau = 1$, integration gives

$$\mathcal{M}_{8/125}\begin{pmatrix} U_d \\ V_d \end{pmatrix} \le \begin{pmatrix} 31/1000 \\ 3/80 \end{pmatrix}, \qquad U_d \le \frac{55813825}{829815232} < \frac{7}{100}, \quad V_d \le \frac{128068875}{3319260928} < \frac{1}{25}.$$

At the zero $s\int_0^T \dot\varphi\,\mathrm{d}\eta = 1/50$ and $\tau < 1/5$; thus

$$K_1 > \frac{1}{10} - \frac{21}{100}\left[\frac{13}{500}\frac{7}{100} + \frac{48}{11} Z_* \frac{1}{25}\right].$$

Let $\Gamma(T;\tau) = F_1(T, \theta^*(T,\tau);\tau)$. Eliminating the radial variation using $F_2 = 0$, and using $|K_2| < 7/100$, yields

$$\begin{aligned} \Gamma_T + \Gamma_\tau &= K_1 - \frac{D_{1\theta}}{D_{2\theta}} K_2 \\ &> \frac{1}{10} - \frac{21}{100}\left[\frac{13}{500}\frac{7}{100} + \frac{48}{11} Z_* \frac{1}{25}\right] \\ &\quad - \frac{7}{500}\frac{7}{100} = \frac{167669417}{1705000000} > \frac{49}{500}. \end{aligned}$$

The T-column at a zero. Here $\ell_T = 0$, $|u(T)| < 19/1000$ and $|P(T)| < 3/80$. The same comparison gives

$$\mathcal{M}_{8/125}\begin{pmatrix} U_T \\ V_T \end{pmatrix} \le \begin{pmatrix} 19/1000 \\ 3/80 \end{pmatrix}, \qquad U_T \le \frac{45639925}{829815232} < \frac{3}{50}, \quad V_T \le \frac{127722375}{3319260928} < \frac{1}{25}.$$

Also

$$\int_0^T |u'|\,\mathrm{d}\eta \le T_* \left[\frac{8}{125} U_T + \frac{48}{11} V_T\right] \le \frac{3734179449}{103726904000} < \frac{21}{500}.$$

Since $s\dot\varphi(T) < \sqrt{b_H} + |k_H| < 7/50$, the angular identity yields

$$\Gamma_T < \frac{7}{50} + C_* \frac{21}{100}\frac{3}{50} + Z_* \frac{21}{500} + \frac{7}{500}\frac{3}{50} = \frac{1370932129}{9687500000} < \frac{71}{500},$$

while $\Gamma_T > 91/2500$ by Theorem 30.

One change. Differentiating $\Gamma(T_s(\tau);\tau)=0$ gives $\Theta_s'=T_s'-1=-(\Gamma_T+\Gamma_\tau)/\Gamma_T<-49/71$ at every zero. Each zero is isolated and is a downward crossing. The zero set is a closed subset of $[\tau_{\mathrm{no}},1/5]$, with neither endpoint a zero. An accumulation point would itself be a zero, contradicting the nonzero derivative there; there are therefore finitely many zeros. If two were consecutive, Θ_s would be negative just after the first and positive just before the second, forcing an intervening zero by continuity. Hence there is exactly one zero. The signs in Proposition XII.9 then give $\Theta_s>0$ before it and $\Theta_s<0$ after it.

Enclosures. At the zero, $1/50=\tau\int_0^1 s\dot\varphi\,\mathrm{d}\nu$. Put $w(r,A)=\sqrt{1-A^2f(r)/2}/\sqrt{h_\varphi(r)}$, $b_\beta=1/(r-2)$ and $t_H=a_Hp_r^2/(b_H\xi^2)$. The purely angular speed is $v_s=\sqrt{b_H}-sk_H=w/(1-sb_\beta w)$, whereas $s\dot\varphi=\sqrt{b_H}/\sqrt{1+t_H}-sk_H$. Since $t_H<1/500$ and $1-(1+t_H)^{-1/2}\le t_H/2$,

$$0\le v_s-s\dot\varphi<\frac{67}{500000}=:e.$$

Set $r_-=1799/300$, $r_+=1801/300$. The function w decreases in r and is concave in A^2. Using $\int_0^1\alpha(\nu)^2\,\mathrm{d}\nu=19/32$, the chord at r_+ and Jensen at r_- give

$$\frac{13}{24}w(r_+,1/2)+\frac{11}{24}w(r_+,1)\le\bar w:=\int_0^1 w\,\mathrm{d}\nu\le\frac{\sqrt{1-19f(r_-)/64}}{\sqrt{h_\varphi(r_-)}}.$$

Rational comparisons after squaring positive quantities give $w(r_+,1/2)>12956/100000$, $w(r_+,1)>11048/100000$, and the rightmost bound is $<1213/10000$. Thus

$$L_w:=\frac{151}{1250}<\bar w<\frac{1213}{10000}=:U_w,\qquad 0<w<\frac{13}{100}=:w_*,$$
$$b_-:=\frac{249}{1000}<b_\beta<\frac{251}{1000}=:b_+.$$

Here $1-b_+w_*>0$. For $s=+1$, Jensen applies to the increasing convex function $w/(1-b_-w)$ for the lower bound, while the upper bound uses the pointwise denominator $1-b_+w_*$. For $s=-1$, the lower bound uses $1+b_+w_*$ and Jensen applies to the increasing concave function $w/(1+b_-w)$ for the upper bound. Including the radial defect e, this gives

$$L_+:=\frac{L_w}{1-b_-L_w}-e<\int_0^1\dot\varphi_+\,\mathrm{d}\nu<\frac{U_w}{1-b_+w_*}=:U_+,$$

$$L_-:=\frac{L_w}{1+b_+w_*}-e<\int_0^1(-\dot\varphi_-)\,\mathrm{d}\nu<\frac{U_w}{1+b_-U_w}=:U_-.$$

Both lower bounds are positive. With $\delta=1/50$, $\delta/U_s<\tau_s^*<\delta/L_s$, namely

$$\frac{96737}{606500}<\tau_+^*<\frac{12124010000}{75418769133},\qquad\frac{10302037}{60650000}<\tau_-^*<\frac{1032630000}{6033081379}.$$

These imply the stated, slightly wider enclosures. □

S11. CONSTANTS OF THE VAIDYA EXAMPLE

Data: $\hat E=\sqrt{2}$, $r_0=r_f=6$, $\delta=\pi/2$, $m_0=1$, $m_1=3/2$ (Remark XIII.9). The bounds needed below are verified in two independent ways, both recorded with the scripts in the accompanying repository: monotone Riemann sums with 4096 or 16384 intervals, with square roots enclosed between consecutive multiples of 10^{-30} and exact rational end points, and cellwise outward-rounded interval enclosures after the substitution $r=t+(4-t)\sin^2\theta$. The displayed enclosures are those of the rational sums.

a. Late accretion, $D = 91$. With $F_a(r) = \sqrt{r(r+a)} - a\,\mathrm{arsinh}\sqrt{r/a}$ one has $F_a'(r) = \sqrt{r/(r+a)}$, so every radial weight of the example is elementary: at mass 1, $R(r) = F_2(r) - F_2(2)$, and $\Lambda(s) = F_2(6) - F_2(s) + F_3(6) - F_3(s)$, whence $\Lambda(11/5) = 5.94120\ldots$ and $2R(6) = 6.45721\ldots$. In the frozen problem at mass 1 the cost metric of the plane is $\mathrm{d}\ell^2 = R'^2\mathrm{d}r^2 + \rho^2\mathrm{d}\varphi^2$ with $\rho^2 = r^2(r-2)/(r+2)$, and $R'/\rho = 1/\sqrt{r(r-2)} = \Phi'(r)$ for $\Phi(r) = 2\,\mathrm{arcosh}\sqrt{r/2}$. A curve that keeps a constant acute angle θ with the radial direction of travel on a band of radii, with φ increasing on both legs, therefore has elementary increments, $\mathrm{d}\varphi = \tan\theta\,|\mathrm{d}\Phi|$ and $\mathrm{d}\ell = \sec\theta\,|\mathrm{d}F_2|$. Take the bands with end points $6 > \frac{527}{100} > \frac{19}{4} > \frac{443}{100} > \frac{433}{100}$ and $\tan\theta = \frac{26}{25}, \frac{38}{25}, \frac{127}{50}, \frac{289}{50}$ on them, from the outermost inwards; descend through them and return symmetrically. The swept angle is $2\sum \tan\theta_i\,[\Phi(b_{i-1}) - \Phi(b_i)] = 1.57395\ldots \geq \pi/2$ and the length is $2\sum \sec\theta_i\,[F_2(b_{i-1}) - F_2(b_i)] = 5.83891\ldots$. The cost depends only on the radial profile (Lemma XI.1): a rail with the same timed radial profile follows the short arc to the target and spends the excess angle on a back-and-forth arc, so it reaches the target exactly, with the same cost, and the frozen infimum is at most 5.8390. The corners at the band ends are admissible for Lipschitz rails. It is below $\Lambda(11/5)$ and below the escape price $2R(6)$; the completion argument and the single-maximum theorem at $\hat{E}^2 = 2$ give the unique minimizer γ_0, of cost $C_0 < 5.8390$. A frozen minimizer reaching $r \leq 11/5$ would cost at least $\Lambda(11/5) > C_0$; so $f > 1/11$ along γ_0 and, the radial end points being equal, $C_0 = \int_0^{T_0} f/\sqrt{2}\,\mathrm{d}v > T_0/(11\sqrt{2})$, whence $T_0 < 11\sqrt{2}\,C_0 < 91$. Both terms in the definition of D are below 91. The strict bounds $1.57395 > \pi/2$ and $5.8390 < 91/(11\sqrt{2}) < \Lambda(11/5) < 2R(6)$ follow from the explicit elementary expressions; rational square-root bounds and the logarithm series with a bounded remainder provide a reproducible verification. The displayed decimal approximations serve only to report their sizes.

b. Critical separation at mass 3/2. By scaling, take $M = 1$, $r_0 = 4$, where $\rho^2 = r^2(r-2)/(r+2)$ and $R' = \sqrt{r/(r+2)}$. With $L = \rho(t)$, $\rho(r)^2 - L^2 = (r-t)\,[(t+2)r^2 + (t^2-4)r + 2t^2 - 4t]/((r+2)(t+2))$, and the angle differential is $t\sqrt{t-2}\,(r+2)\,\mathrm{d}r/\sqrt{H_t(r)}$ with $H_t = r(r-2)(r+2)(r-t)[(t+2)r^2 + (t^2-4)r + 2t^2 - 4t]$, whose discriminant $2^{24}t^{10}(t-2)^5(t+2)^3(t^2-8t-4)(t^2+2t-4)^2$ does not vanish for the periastra used below: the differential lives on a smooth genus-two hyperelliptic curve. No elementary or elliptic reduction is used here; the following enclosures are obtained from monotone Riemann sums with rigorous error bounds. With the periastra $t_a = 29/10$ and $t_b = 297/100$, the substitution $r = t + (4-t)u^2$ removes the singularity of the dip integrals, the angle integrand decreases and the length integrand increases in u, and

$$\Delta(t_a) \in (1.605672, 1.605738), \qquad \Delta(t_b) \in (1.558655, 1.558714),$$
$$\ell(t_b) \in (3.091180, 3.091274), \qquad 2R(4) \in (3.081722, 3.081750).$$

Thus $\Delta(t_a) > 39/25 > \Delta(t_b)$ and $\ell(t_b) > 2R(4)$. Since $t_a < t_b$ and $\Delta(t_a) > \Delta(t_b)$, the single maximum of the dip angle puts t_b on the decreasing, shallow branch, the shallow dip of angle $39/25$ has its periastron between t_a and t_b and is longer than the dive, and the deep dip is longer as well; hence $\delta_c < 39/25 < \pi/2$.

c. Early window, $h_* = 1/1000$. Let a minimizer exist for a profile equal to $3/2$ after $h \leq 1/1000$. Since $|\dot{r}| < 4$ and $|\dot{\varphi}| \leq \sqrt{2}/r$, the layer keeps $r \in [5,7]$ and sweeps at most $3h/10$; the layer alone cannot reach the target, so $T > h$, and the frozen tail from (r_h, φ_h) must sweep $\delta_h \geq \pi/2 - 3h/10$ while costing at most its own escape price, since by the principle of optimality it is a frozen minimizer from (r_h, φ_h). A monotone frozen geodesic between radii in $[5,7]$ sweeps less than $\arccos\sqrt{125/392} < \pi/3$, so the tail is a dip with Clairaut constant J. With $r_t = \rho^{-1}(J)$,

$$\mathcal{S}_J(a,b) = \int_{r_t}^{a} \sqrt{1 - J^2/\rho^2}\,R'\,\mathrm{d}r + \int_{r_t}^{b} \sqrt{1 - J^2/\rho^2}\,R'\,\mathrm{d}r,$$
$$F_J(a,b) = \frac{R(a) + R(b) - \mathcal{S}_J(a,b)}{J},$$

the length of the dip is $\mathcal{S}_J + J\delta_h$, and since $J \le \min\{\rho(r_h), \rho(6)\}$ every radius between r_h and 6 is at least r_t. The cost bound reads $\delta_h \le F_J(r_h, 6)$, while $\partial_a F_J = R'(a)\big(1 - \sqrt{1 - J^2/\rho(a)^2}\big)/J \in [0, R'(a)/\rho(a)]$, by $1-\sqrt{1-x} \le x$ and $J \le \rho(a)$, including the tangent case by continuity, and $R'/\rho = 1/(a\sqrt{f(a)}) < 1/3$ on $[5, 7]$ and $F_J(6, 6) \le \delta_c < 39/25$ by the Maxwell rule. Hence $\delta_h < 39/25 + 4h/3$, against $\delta_h > 157/100 - 3h/10$; at $h = 1/1000$ the gap is $251/30000$.

d. The translated step. For the cubic step of duration $\epsilon = 1/2000$, every $d \le 1/2000$ ends the accretion by $1/1000$, and Theorem 38 gives $1/2000 < d_* < 91$.